

Advancements in Noninvasive Blood Pressure Measurement: A Comprehensive Review of Cuffed, Cuffless, and Contactless Methods

Terezie Kauzlaricova¹, Martin Augustynek², and Jan Kubicek³

Abstract—Recent advances in noninvasive blood pressure (BP) monitoring are transforming the field, offering alternatives to traditional cuff-based methods that are often limited by discomfort and intermittent measurement. Following PRISMA guidelines, this review systematically analyses studies published between 2020 and 2024, categorizing current BP measurement techniques into cuff-based, cuffless, and contactless methods. Special attention is given to emerging technologies such as photoplethysmography (PPG), electrocardiography (ECG), bioimpedance (BIOZ), radar sensing, and remote PPG (rPPG), as well as the integration of artificial intelligence (AI) for signal processing and BP estimation. This review evaluates the comparative performance, clinical relevance, and readiness for real-world application across modalities. While cuffless and contactless approaches show significant promise for continuous and user-friendly monitoring, their accuracy often depends on calibration, multimodal data fusion, and robust algorithm design. Ethical considerations related to unobtrusive monitoring are also discussed. This work highlights current limitations, identifies future research opportunities, and supports the development of next-generation, AI-driven BP monitoring systems suitable for both clinical and consumer use.

Index Terms—Artificial intelligence (AI), biomedical measurement, biomedical monitoring, blood pressure (BP) measurement, machine learning (ML), noninvasive measurement, wearable health monitoring.

I. INTRODUCTION

BLOOD pressure (BP) is a critical physiological parameter that reflects the force exerted by circulating blood on the walls of blood vessels. It is typically expressed in millimeters of mercury (mmHg) and is represented by two values: systolic BP (SBP), which measures the pressure during heartbeats, and diastolic BP (DBP), which measures the pressure between heartbeats. Accurate measurement of BP is essential for diagnosing and managing hypertension, a condition that significantly increases the risk of cardiovascular diseases, stroke, and renal failure [1], [2]. The World Health Organization has

reported that hypertension affects approximately 24% of men and 20.5% of women globally, highlighting its prevalence and the need for effective monitoring [2].

Accurate BP readings are vital to identifying and preventing a range of chronic diseases. For instance, research utilizing machine learning (ML) models for kidney disease prediction has underscored the importance of tracking systolic and DBP as a predictor of renal failure [3], [4]. Inaccurate readings can result in misdiagnosis and inappropriate treatment, either by falsely classifying patients as hypertensive or by underestimating their condition [5], [6]. For instance, studies have shown that an overestimation of SBP by just a few mmHg can significantly increase the number of patients diagnosed with hypertension, leading to unnecessary treatment [5], [6]. Conversely, underestimation can result in untreated hypertension, which poses serious health risks. The American Heart Association and other health organizations advocate for the use of automated and home BP monitoring to mitigate the white coat effect, where patients exhibit elevated BP readings in clinical settings due to anxiety [5], [7], [8].

Neglecting to monitor and manage BP can lead to severe health consequences. Hypertension is a major risk factor for various cardiovascular diseases, including coronary artery disease, heart failure, and stroke [1], [9]. Furthermore, conditions such as isolated diastolic hypotension have been associated with increased mortality, which emphasizes the value of both high and low BP readings [10]. Chronic hypertension can lead to end-organ damage, affecting the kidneys, eyes, and brain and can significantly impact quality of life [1], [11]. Additionally, sleep disorders have been linked to elevated BP, indicating that lifestyle factors also play a role in hypertension management [11], [12].

In summary, BP is a vital sign that needs to be measured accurately to ensure the proper diagnosis and treatment of hypertension. The implications of neglecting BP monitoring are profound, leading to increased morbidity and mortality associated with cardiovascular diseases. Accurate measurement techniques, including automated and home monitoring, are essential to provide reliable data for effective health management [5], [7], [8].

This article provides an in-depth examination of the latest scientific research focused on the development of techniques for measuring BP. While previous reviews have focused on specific aspects, such as automated auscultatory techniques [13], video-based approaches [14], continuous monitoring [15], [16], or artificial intelligence (AI)-driven estimations

Received 20 October 2025; revised 23 November 2025; accepted 7 December 2025. Date of publication 2 January 2026; date of current version 16 January 2026. This work was supported in part by European Union through the LERCO Project via the Operational Programme Just Transition under Grant CZ.10.03.01/00/22_003/0000003 and in part by the Biomedical Engineering Systems XXI under Project SP2025/032. The Associate Editor coordinating the review process was Dr. Rui Yang. (Corresponding author: Terezie Kauzlaricova.)

The authors are with the Department of Cybernetics and Biomedical Engineering, VSB–Technical University of Ostrava, 708 00 Ostrava, Czech Republic (e-mail: terezie.kauzlaricova@vsb.cz).

This article has supplementary downloadable material available at <https://doi.org/10.1109/TIM.2026.3650271>, provided by the authors.

Digital Object Identifier 10.1109/TIM.2025.3650271

[17], [18], this study aims to distinguish itself by presenting a broader spectrum of measurement methods and incorporating more recent studies. The primary objective is to evaluate the advantages and limitations of traditional, cuffless, and contactless methods for measuring BP, with a particular focus on their accuracy, clinical applicability, and integration with AI. The study explores whether noninvasive methods, particularly those utilizing ML and multimodal signal processing, can achieve accuracy comparable to conventional cuff-based approaches while offering improved convenience and continuous monitoring capabilities. It reviews contemporary techniques for assessing BP, focusing on noninvasive strategies, and investigates the processing of these measurements to enhance BP management. To further clarify these differences, this review includes dedicated summary tables that contrast its scope and methodology with previous surveys, thereby highlighting its unique contribution.

The composition is organized in the following manner. Section II deals with the organization of the review concerning technical techniques for measuring blood. Section III focuses on the description of methods for measuring BP based on their measurement principles, including cuff-based, cuffless, and contactless techniques. Section IV examines techniques for estimating BP, either through deterministic approaches or using AI. Section V presents a comparison of study effectiveness based on signal acquisition methods and signal processing techniques. All abbreviations used throughout this article are summarized in Table S1 (Supplementary Material).

II. STRUCTURE OF THE REVIEW

The articles and studies chosen for this review span from January 2020 to February 2024, following the PRISMA methodology. In addition, selected relevant articles published in early 2025 were included to reflect the most recent developments in the field. The articles were chosen based on keywords: *[BP AND measurement AND (technique OR methods) AND noninvasive]*. Research was conducted in the databases EBSCO and IEEE Xplore. The exploration within chosen databases was conducted following the PRISMA review strategy, as illustrated in Fig. 1. This review was conducted by utilizing comprehensive documents, encompassing original research articles, conference papers, theses, book chapters, and review articles.

The quality of individual studies was not considered, particularly regarding the inclusion of a new approach to BP measurement that encompasses both hardware and software solutions. We also reviewed studies that did not include a final assessment of the quality of their solution and were therefore only in the experimental or early testing phase. We focused on studies focusing on noninvasive methods of BP measurement. Studies that focused on invasive methods were excluded. In addition to other search criteria, we exclusively focused on papers written in the English language.

A total of 1142 articles were initially gathered from various academic databases, ensuring a wide range of sources for a thorough review. Out of these, six articles were excluded right away, as they were written in a foreign language that fell outside the scope of our analysis. This left us with 1136 articles for further evaluation. The first round

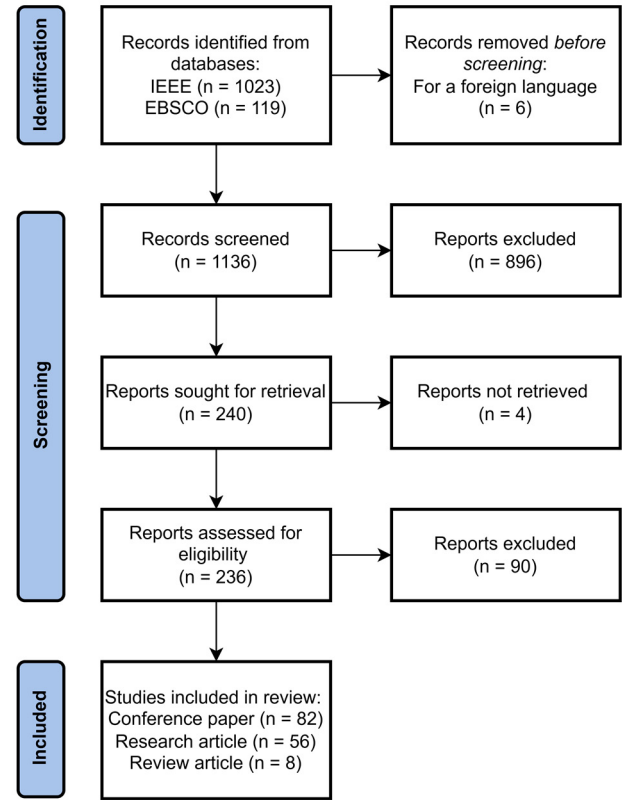


Fig. 1. PRISMA flow diagram of the review structure.

of screening involved reviewing the titles of each article to determine their relevance to the core topic of the review. At this stage, 896 articles were set aside because their titles did not sufficiently align with the specific focus of the study.

After conducting an initial screening, we identified 240 articles that appeared more closely related to the subject matter. Unfortunately, four of these articles could not be accessed due to technical limitations or restrictions on the hosting databases. This left a total of 236 articles available for more profound analysis.

During the next phase of review, we carefully examined these 236 articles. However, 90 of them were excluded because they were either duplicates, included instructions for BP measurement, assessed long-term BP measurement, described the principles of BP measurement, or were deemed irrelevant after a more detailed assessment of their content. This resulted in a final selection of 146 articles, of which 82 were conference papers, 56 were research articles, and 8 were review articles, all of which were subjected to a comprehensive and detailed review to ensure a robust and well-rounded understanding of the topic. These 146 articles form the backbone of our analysis and contribute significantly to the findings presented in this review.

III. METHODS BY MEASUREMENT: CUFF-BASED, CUFFLESS, AND CONTACTLESS

This section categorizes noninvasive BP measurement techniques into three main groups based on their measurement

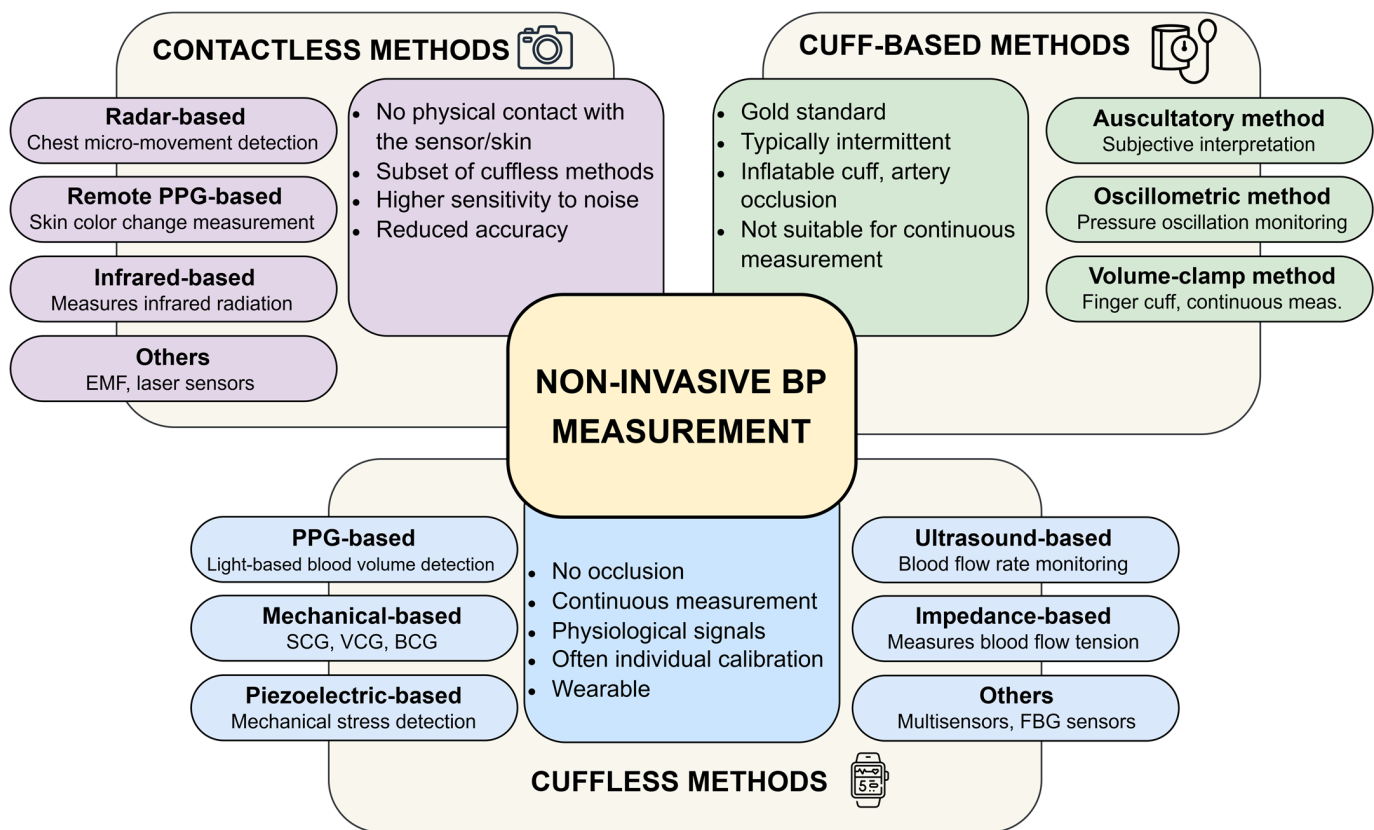


Fig. 2. Mind map of noninvasive BP measurement methods, categorized into cuff-based, cuffless, and contactless approaches.

approach and physical interaction with the subject: cuff-based, cuffless, and contactless methods. A schematic mind map summarizing this classification is presented in Fig. 2 to provide an overview of the methodological landscape.

Cuff-based methods rely on an inflatable cuff to occlude the artery, typically on the upper arm or wrist. These methods estimate systolic and diastolic pressure based on auscultatory or oscillometric signals during cuff inflation or deflation. While clinically established and accurate, they are typically intermittent and not suitable for continuous monitoring.

Cuffless methods do not require vascular occlusion. Instead, they estimate BP through physiological signals such as photoplethysmography (PPG), electrocardiography (ECG), bioimpedance (BIOZ), or mechanical vibrations measured via wearable or embedded sensors. These techniques often use pulse transit time (PTT) or signal morphology features combined with regression models or AI-based algorithms. Cuffless methods enable continuous, wearable, and user-friendly monitoring but often require individual calibration.

Contactless methods are a subset of cuffless techniques that do not require any physical contact with the body. They rely on remote sensors, such as radar, cameras, or infrared (IR) imaging, to capture physiological activity (e.g., chest wall motions, skin color changes, or thermal patterns) from a distance. These signals are then processed to extract pulse wave characteristics or estimate PTT indirectly. While highly convenient and unobtrusive, contactless methods are currently more prone to signal noise, environmental interference, and reduced accuracy.

It is important to note the distinction between cuffless and contactless methods. Cuffless techniques rely on physiological signals acquired through sensors in direct contact with the body (e.g., PPG, ECG, piezoelectric, or ultrasound sensors), but without an inflatable cuff. In contrast, contactless approaches operate without any physical sensor-skin interface, instead relying on remote modalities such as camera-based PPG, IR imaging, or radar sensing.

Fig. 3 shows that most of the current studies are based on the cuffless method. Utilizing cuffless and contactless techniques provides the benefit of ongoing measurement; however, this often comes with a degree of measurement uncertainty.

A. Cuff-Based Methods

The two primary approaches frequently employed by outpatient clinics are outlined here. Methods of oscillometric and auscultation. Both approaches can be evaluated based on their original proposals, while innovative procedures and techniques are currently being developed to assess these established methods. There are additional methods that may not be as widely recognized, such as the volume clamp technique, sometimes referred to as Peña's method, which involves the use of finger clamps.

Cuff-based methods remain the gold standard for BP measurement, providing reliable and accurate readings in both clinical and home settings. These methods involve the use of an inflatable cuff, typically wrapped around the upper arm, which temporarily occludes the brachial artery. As the

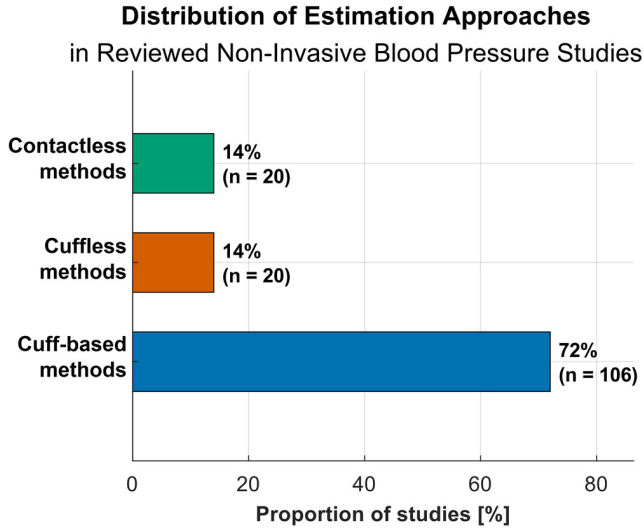


Fig. 3. Distribution of types of BP measurement in this review.

cuff inflates, it restricts blood flow, and once the pressure is gradually released, blood begins to flow back through the artery. This process is monitored using either auscultatory or oscillometric techniques to estimate systolic and DBP values.

1) *Auscultatory Method*: The auscultatory method of BP measurement is a traditional and still widely accepted technique for noninvasive BP measurement [13], [19]. This method relies on the detection of Korotkoff sounds as the pressure in the arm cuff is slowly lowered. These sounds show blood flowing through the brachial artery. During the measurement, the cuff on the arm inflates to a pressure higher than the expected systolic pressure, completely stopping the flow of blood through the brachial artery. Thereafter, the pressure in the cuff gradually decreases [20]. Once the pressure drops below the patient's systolic pressure, blood begins to flow through the artery, producing sounds known as Korotkov sounds. The first audible sound is interpreted as systolic pressure, while the last audible sound (when the artery remains permanently open) is considered diastolic pressure. Fig. 4 describes the process of obtaining pressure by auscultation.

This method requires experience and expertise, because the moment when Korotkov's sounds are heard can be subjectively interpreted. This can lead to errors and inaccuracies, especially when measurements are made by less experienced personnel or when borderline values are interpreted [20], [21]. This subjectivity can lead to incorrect diagnosis or assessment of the patient's cardiovascular health. The method has some inherent uncertainty associated with the variability of measurements between individuals and instruments. This uncertainty stems from the very nature of the method and the physiological factors that affect the blood flow and audibility of sounds [21].

Alternative approaches to measurement focus on automation. This is the use of audio or amplitude analysis and estimation based on AI [13], [22]. Furthermore, algorithms use the period and frequency of Korotkov sounds, where these parameters can provide greater accuracy in automated instruments that traditionally use oscillometric methods [13], [23]. Emphasis is also placed on using the amplitude of

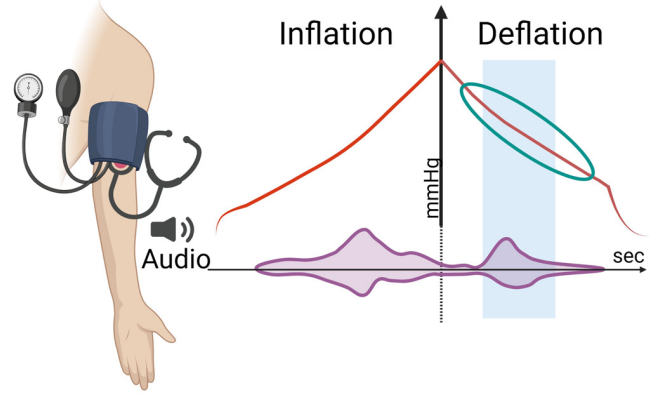


Fig. 4. Principle of cuff-based BP measurement. An inflatable cuff is placed on the upper arm and gradually deflated at a constant rate. As blood flow resumes in the artery, sounds known as Korotkoff sounds can be detected with a microphone. The first sound indicates SBP, while the disappearance of sounds marks DBP. The red trace shows cuff pressure, the blue curve represents the oscillometric envelope, and the purple line depicts the recorded Korotkoff sounds. The highlighted region illustrates the interval between SBP and DBP. Adapted from [20].

Korotkov sounds as a threshold criterion to improve sound detection and reduce measurement errors [19].

2) *Oscillometric Method*: The oscillometric method for BP measurement is commonly employed in automated devices for noninvasive BP assessment. Its fundamental principle lies in monitoring pressure oscillations during the gradual release of pressure in the cuff [24], [25]. Various factors, including inflation speed, cuff tightness, and heart rate, can influence measurement accuracy [26]. In addition to the arm, the new methods also use the wrist, where they use a device to detect the oscillogram from the radial artery [27], [28].

One potential innovation for enhancing the algorithm used in BP calculation is the application of correction curves based on a Gaussian distribution. This approach aims to eliminate artifacts and noise that may distort an oscillometric signal [29]. Another approach estimates the BP value by AI [30], [31].

Alternative approaches to oscillometric measurement include techniques that initiate measurements during cuff inflation [32]. In this method, the device gradually inflates the cuff while simultaneously measuring oscillations to identify the diastolic pressure first, followed by the systolic pressure. Once these values are determined, the cuff is quickly deflated.

The oscillometric method is well-suited for home monitoring, as it does not require the expertise of a trained healthcare professional, unlike the auscultatory method. It is thus an appropriate choice for telemedicine applications, where devices can transmit measured BP values directly to healthcare providers for remote monitoring [33].

3) *Volume-Clamp Method*: In addition to these conventional methods, more advanced techniques like the volume-clamp method and the volume-compensation method offer alternatives that enhance continuous BP monitoring. The volume-clamp method, also known as the Penaz method, involves placing a cuff on a finger rather than the upper arm. This method works by continuously adjusting the pressure in the cuff to maintain a constant blood volume in the finger's artery, which is measured using PPG. By dynamically controlling the cuff pressure to keep the arterial volume stable,

this method provides real-time, beat-to-beat BP readings. The volume-clamp approach is particularly useful in continuous monitoring, such as during surgery or in intensive care, but it can be more susceptible to movement artifacts and less accurate during vasoconstriction or in patients with poor peripheral circulation [34], [35].

B. Cuffless Methods

Cuffless techniques for assessing BP encompass a myriad of diverse measurement approaches. Most of these methods can be measured continuously. Undoubtedly, the primary approach currently under thorough examination by numerous researchers involves the utilization of PPG. In addition, these are methods based on the extraction of the signal from the mechanical properties of the heart or they detect by using BIOZ, piezoelectric, or optical sensors.

1) *PPG-Based Methods*: Assessing BP through the PPG technique might seem straightforward in theory, but in practice, it presents a range of challenges. The fundamental concept of PPG, which involves detecting changes in blood volume using light, is well understood. However, developing a method that provides precise and reliable BP readings requires more than just basic knowledge [36]. It involves intricate mathematical equations to interpret the data, a profound understanding of the physiological mechanisms governing blood flow and pressure, and the use of highly sensitive and accurate sensors. In some cases, multiple devices are employed to ensure consistent data acquisition, which further complicates implementation [37].

The PPG method relies on the interaction between a light-emitting diode (LED) or IR diode and the patient's skin. Specifically, the LED emits light that is either transmitted through or reflected from the skin. The basic places where the PPG signal is collected on the body are the finger, preferably the index finger on the left hand [38], [39], [40]; the wrist, where it is measured, for example, with a smartwatch [41], [42], [43] or tightening system [44]; or on the chest with so-called chest bands.

Changes in light absorption or reflection correspond to pulsatile blood volume variations with each heartbeat [45]. This waveform can provide cardiovascular metrics, such as heart rate and, with additional modeling, BP. Achieving accurate results requires sophisticated processing to account for factors like arterial stiffness, which varies across individuals. For this reason, most studies using only PPG to estimate BP incorporate AI models [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65].

Because deriving BP from PPG alone is difficult [66], [67], the signal is often combined with ECG to improve robustness. One of the most common strategies is to pair PPG with ECG using Einthoven leads, a standard configuration in clinical practice [56], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78]. This enables precise detection of R peaks, which serve as reference points for cardiovascular cycles. In some cases, a phonocardiogram (PCG) can be used as an alternative [79].

Based on these landmarks, researchers calculate PTT, the time for the arterial pulse to travel from the heart (detected

by ECG) to the peripheral vasculature (detected by PPG). PTT is a powerful surrogate marker of BP, as it reflects arterial stiffness and hemodynamic changes. This approach has been validated in numerous studies [36], [37], [45], [80], [81], [82] and is widely adopted in wearable and remote monitoring systems. When deterministic models are not applied, PTT features are commonly used as inputs for data-driven algorithms [70], [72], [83], [84], [85], [86], [87], [88], [89], [90], [91].

Another approach involves using a standard light source together with a smartphone camera, where the fingertip is placed directly on the lens. Light variations transmitted through the tissue are recorded as a video sequence, from which the red channel is extracted and converted into a waveform. This signal can then be analyzed with ML models to estimate BP [67], [92], [93], [94], [95]. Although this technique still requires finger contact and is therefore considered cuffless, its underlying principle closely resembles remote PPG (rPPG), which is discussed in greater detail in Section III-C.

2) *Mechanical-Based Methods*: A fundamental approach to evaluating a patient's BP involves observing the mechanical movements, or vibrations, of the chest that arise from cardiac activity and the mechanical properties of the cardiovascular system. One key technique used to capture these vibrations is seismocardiography (SCG). In SCG, highly sensitive sensors, typically accelerometers, are placed on the chest to precisely detect the subtle movements of the chest wall caused by the heart's mechanical action. By analyzing the SCG signals, clinicians can identify specific cardiac events, such as the systolic peak, which corresponds to the contraction of the heart [96]. To enhance the accuracy of BP estimation, SCG is often paired with an ECG for calculating PTT. This combination of SCG and ECG data allows for an indirect, yet effective, estimation of BP [97], [98].

In addition to SCG, research into vibrational cardiography (VCG) has gained traction, particularly for its integration of gyroscopic sensors alongside traditional accelerometers. This advanced combination allows for a more comprehensive capture of the heart's vibrations by measuring both linear and rotational movements of the chest. The inclusion of a gyroscope in SCG recordings transforms the signal into a VCG output, enabling more sophisticated data analysis [99].

Another method that leverages the mechanical properties of the cardiovascular system is ballistocardiography (BCG). BCG measures the body's recoil caused by the force of blood ejection from the heart [100]. This can be done using various configurations, such as a chest strap equipped with sensors that detect the small movements produced by the heart's contractions. Some BCG devices also incorporate additional sensors for ECG or PPG measurements, providing a multimodal approach to monitoring cardiovascular health [100], [101], [102]. An innovative application of BCG includes a footpad sensor, which integrates both BCG and PPG technology, as shown in Fig. 5. In this setup, the footpad detects the BCG signal from the body's micromovements, while the PPG sensor measures blood volume changes at the foot [103]. This novel design demonstrates how BCG can be applied in noninvasive, everyday scenarios, making it a versatile tool for long-term BP monitoring and cardiovascular assessment.

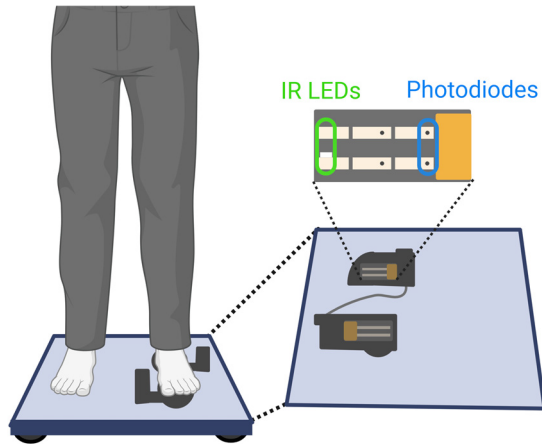


Fig. 5. Combination of BCG and PPG for cuffless BP estimation. A subject stands on a force plate integrated with PPG sensors at the sole of foot. The BCG captures body micromovements caused by cardiac ejection, while the PPG records pulse waves. The time difference between these signals (PTT) can be used to estimate BP. Adapted from [103].

3) *Piezoelectric Methods*: Piezoelectric-based methods represent a noninvasive and increasingly sophisticated approach to measuring BP by detecting mechanical strain or pressure changes on the skin surface caused by arterial pulsations. Piezoelectric-based methods can give continuous and real-time measurements without blocking the artery, which is not possible with traditional cuff-based methods. These methods typically involve placing a piezoelectric-based sensor over a superficial artery, such as the radial [104], [105], [106], [107], and respond to minute deformations in the skin induced by pulsatile blood flow. The piezoelectric material converts these mechanical deformations into electrical signals, which are then processed to estimate systolic and DBP values.

Additionally, these sensors can measure pulse pressure wave (PPW) and signal amplitude, providing more detailed information about cardiovascular health. This method allows for the integration of multiple sensors, specifically two PPW modules and an ECG signal [108]. Each PPW module consists of a piezoelectric sensor, with one module positioned on the foot and the other on the wrist. The signals obtained from these sensors are subsequently used to extract features, which are then processed using regression models to estimate BP [108].

Prototypes of flexible sensors are currently in development, with the potential to be compact and integrated into wearable technology in the near future [104], [105], [106], [107], [109], [110].

4) *Ultrasound-Based Methods*: One of the core principles behind ultrasound-based BP measurement is the ability to track the velocity of blood flow using the Doppler effect [111]. By measuring the shift in frequency of the ultrasound waves reflected by moving blood cells, the system can calculate blood velocity, which, when combined with arterial wall movement data, provides valuable information about BP. Additionally, ultrasound can measure arterial distensibility or how the artery expands and contracts with each heartbeat. This distension is closely related to BP because arteries that are stiffer (which is often a sign of high BP) do not stretch as much when pressure changes, while arteries that are more elastic do. Recent work

has further exploited pulsed-wave Doppler signals from the carotid artery to extract waveform indices for estimating local pulse wave velocity (PWV), offering a calibration-free and pressure-independent indicator of arterial stiffness [112].

Beyond Doppler techniques, ultrasound can assess arterial wall deformation and resonance frequency, which are key indicators of arterial compliance [113]. As BP increases, arterial stiffness typically rises, reducing vessel distension. Measuring these biomechanical properties allows for BP estimation through analytical models or biomechanical simulations, often without the need for traditional calibration. Recent studies have also introduced flexible ultrasound sensors based on piezoelectric composite structures, which provide a high electromechanical coupling factor and enable sensitive detection of arterial wall motion in wearable or skin-conformal formats [109], [110].

5) *Impedance-Based Methods*: Impedance-based methods utilize standard silver [114] or Ag/AgCl [115] electrodes placed at various locations such as the finger [114], wrist [115], or head [116]. These methods measure impedance, which is the change in voltage that happens when blood flows through the area being measured. The acquired signal can be processed in one of two ways: 1) independently, using Ohm's law while assuming that impedance changes are linearly correlated with changes in arterial cross-sectional area [115]; or 2) by extracting features from the signal as inputs for regression models [114], [116].

The BrainZ-BP system [116] exemplifies this approach by incorporating both BIOZ and ECG signals to estimate BP. In this system, two electrodes are positioned on the forehead and occipital bone in an anteroposterior configuration to measure BIOZ, while standard ECG measurements are included. Features such as PTT, pulsewidth, and pulse amplitude are extracted from the signals and used as inputs for regression models [116].

6) *Other Methods*: Cuffless methods could also include BP sensing using a multisensor in the ear, where they use a pressure, differential pressure, and an optical sensor in the ear [117]. Optical sensors are also used in a method that directly targets fiber Bragg grating (FBG) sensors to capture radial arterial pulse pressure waveforms [118], [119].

C. Contactless Methods

Contactless approaches acquire cardiovascular signals without any direct physical sensor-skin interface. Within this category, two major families can be distinguished: 1) optical methods, including rPPG and IR thermography, which rely on light reflection or thermal emission from the skin surface; and 2) radio frequency methods, such as radar and electromagnetic field (EMF) sensing, which capture chest wall or arterial motion at a distance. Although rPPG is based on the same optical principle as contact PPG, it is classified here as contactless because signal acquisition is entirely camera-based and requires no attached sensor. Similarly, radar technologies are included under contactless methods since most reported studies employ them in unobtrusive, remote configurations.

1) *Radar Methods*: Radar-based approaches estimate BP from micromotions on the chest wall or arterial surfaces caused by pulsatile blood flow. These methods operate by

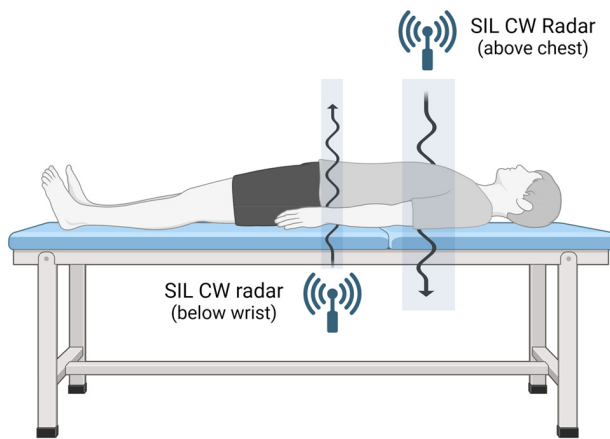


Fig. 6. Contactless radar-based BP monitoring. The experimental setup uses two continuous-wave radars: one aimed at the chest and one placed beneath the wrist through an opening in the bed. By simultaneously capturing pulse signals at both locations, PTT can be estimated without direct contact with the subject. Adapted from [122].

transmitting low-power microwave signals and analyzing the reflected waves to detect periodic displacements corresponding to cardiac activity. Typically, radar systems are aimed at the chest at a distance of 10–20 cm [120], [121], as chest motion provides strong and stable reflections from respiration and cardiac activity. Some systems target the wrist [122], where the arterial pulse is accessible, though the signal is weaker. In more advanced setups, multiple radars may be used, as shown in Fig. 6, with one directed at the left side of the chest and another at the wrist, to capture complementary information and improve robustness [123], [124], [125]. Other configurations have explored self-injection locked (SIL) radar, in which the transmitted oscillator is synchronized with the weak reflected signal from the body to enhance phase sensitivity and reduce phase noise [123]. Frequency-modulated continuous wave (FMCW) radar has also been investigated to assess the impact of measurement geometry, showing that accuracy declines with distances greater than 15 cm or when the radar is positioned at oblique angles exceeding $\pm 60^\circ$ [126].

A commonly used radar type is continuous wave (CW), as it provides a simple, low-cost design with high sensitivity to micromotions such as respiration and pulse waves, making it well-suited for short-range BP monitoring. The Doppler radar is another common way to pick up the pulse wave. It does this by measuring the frequency shift (Doppler effect) between the transmitted and reflected signals [127], [128]. This frequency shift occurs when an object, such as a blood vessel, moves toward or away from the radar. The precise measurement of pulse waves allows for the calculation of PTT, which is often employed in ML models to estimate BP. In cases where PTT is not used as a feature for ML, the raw pulse wave data itself serve as the input.

The frequency bands employed vary considerably. The choice of a radar frequency band strongly influences measurement performance. Lower bands such as 2.4 GHz are attractive due to their low cost and ability to penetrate obstacles, though they provide limited resolution and sensitivity to subtle chest movements [121], [124], [129]. Mid-range frequencies around 24 GHz offer a practical tradeoff, with smaller antennas

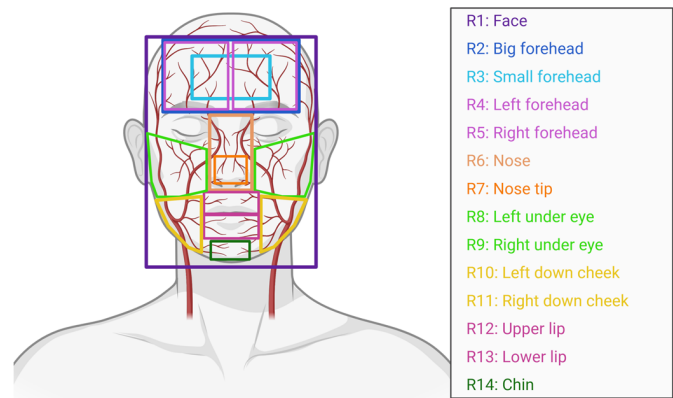


Fig. 7. Regions of interest (ROIs) are used for rPPG. Subtle changes in skin color, caused by variations in blood volume, can be extracted from facial videos to estimate BP. Commonly examined areas include the forehead, nose, and cheeks, which are less affected by motion and lighting conditions. The figure highlights these facial regions where rPPG signals are typically obtained. Adapted from [133].

and improved resolution while maintaining reasonable penetration [120]. Millimeter-wave (mmWave) radars operating at 77–81 GHz provide the highest spatial resolution and sensitivity to micromotions such as pulse waves but suffer from stronger attenuation and limited penetration, making them most suitable for short-range, high-precision applications [122]. This variability in frequency choice highlights the tradeoffs between range, resolution, hardware complexity, and environmental robustness.

The most frequently applied ML methods for determining BP from radar-based measurements are random forest (RF) models, or their variants such as regression RF [120], [121], [124], [128], as well as neural networks (NNs) [122], [123]. RF models offer robustness in handling nonlinear relationships between input features and BP, while NNs, particularly deep learning (DL) architectures, have demonstrated the ability to model more complex physiological processes, making them suitable for radar-based BP estimation.

2) *Remote PPG*: Alongside radar, technologies such as image PPG (iPPG), often referred to as rPPG or video-based PPG (vPPG), are also utilized. In rPPG measurement, facial video is most commonly utilized [130], [131], [132], [133], [134] or, alternatively, videos of the skin where the specific region of measurement on the body can be hand [135] or not specified in [136]. A camera or smartphone camera [137] is typically positioned approximately 30–50 cm in front of the subject [138]. This proximity ensures the capture of subtle variations in skin color caused by changes in blood volume, which are key to extracting the pulsatile signal. From the recorded video, the signal is extracted from specific facial regions (Fig. 7), such as the forehead, nose, or cheeks, areas known for having relatively stable illumination and minimal motion artifacts [138], [139]. Various methods are available for extracting the signal, including the averages of green color intensity, as green light is most absorbed by hemoglobin, providing a stronger pulse signal; the optimal linear combination of color channels using independent component analysis (ICA), which separates the source signals from mixed observations; a chrominance-based (CHORM) signal standardized

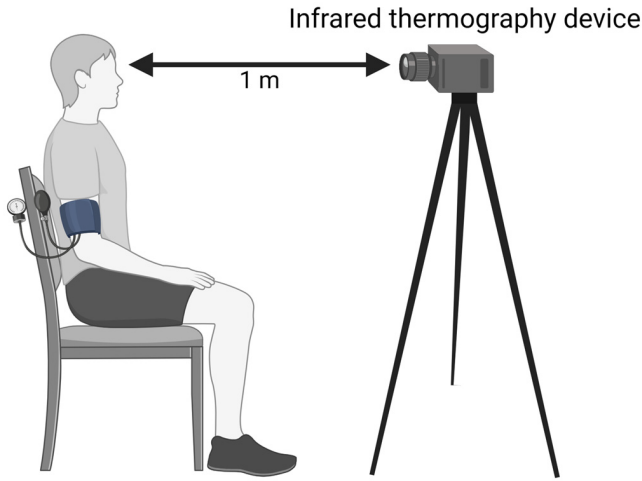


Fig. 8. IR camera system for contactless BP estimation. An IR camera records thermal variations in the hand or face, reflecting blood flow and vascular activity. These signals can be processed to estimate BP without physical contact. Adapted from [143].

by the skin color profile, which enhances the pulse signal by minimizing lighting effects; and the plane orthogonal to skin tone method, which projects color variations onto a plane orthogonal to the skin tone to isolate the blood volume pulse [135], [140]. More recent work further extended video-based monitoring by combining multisite and multiwavelength rPPG signals into multidimensional PTT features [141].

After signal extraction using one of these methods or from predefined landmarks, the signal undergoes preprocessing through filtering to remove noise, motion artifacts, and ambient lighting interference, ensuring more accurate pulse detection [135], [137]. One approach for estimating BP from rPPG is to calculate PTT from the waveform, specifically from its peaks, which correlates with BP changes, allowing it to be used for further analysis and estimation of cardiovascular metrics [135]. Alternatively, ML techniques have been increasingly employed to enhance the precision of rPPG-based measurements, with authors utilizing models such as deep NNs (DNNs), DL, long short-term memory (LSTM) networks for time-series analysis, or convolutional NNs (CNNs), which excel in pattern recognition [130], [132], [136], [137], [138]. These models are trained on features extracted from the rPPG signal, along with patient demographic data such as age, gender, skin tone, and physiological parameters. This integration of ML allows for the development of personalized models capable of accounting for individual variability in signal characteristics, improving the accuracy of BP estimation and other physiological measurements.

3) *IR Camera*: Methods based on IR cameras operate on principles like rPPG, with the key difference being that IR cameras capture IR radiation rather than visible light, allowing them to measure the temperature at the target site. Common measurement locations include the hand [142] and the face [143], [144], with distances up to 1 m from the camera (Fig. 8).

Thermal images captured by IR cameras are often processed using algorithms such as ICA, which decomposes the signal into independent components to generate features [143], or by employing advanced DL models like DenseNet201 for

feature extraction [142]. The extracted features are then used as inputs for AI methods [134], [142], [143], [145] to estimate physiological parameters, including BP.

In addition to standard IR cameras, near-IR (NIR) cameras are also employed for similar purposes [146]. NIR cameras measure parameters such as heart rate and oxygen saturation, from which blood flow values are derived. These values are then used to estimate BP. The use of NIR cameras offers significant advantages, particularly in overcoming the limitations of reduced resolution in contactless measurement techniques based on image sequences [146].

4) *Other Methods*: Among the advanced methods for determining BP values is the use of EMFs [147]. This technique involves a device that detects the voltage or EMF generated by the flow of blood within an artery. According to Faraday's law of electromagnetic induction, the velocity of blood flow can be calculated using this EMF. To obtain BP values, the mean arterial pressure (MAP) is first calculated using the flow rate and total peripheral resistance [147].

Additionally, laser displacement sensors can be employed to estimate BP [148]. These sensors are directed at the chest and abdomen, measuring body displacement in multiple positions, which allows for the monitoring of the propagation of the cardiac pulse wave. By comparing the cardiac pulse wave from both sensors, the relationship between PTT and BP is evaluated [148].

IV. SIGNAL PROCESSING AND BP ESTIMATION TECHNIQUES

While Section III focused on the physical principles and sensor modalities for acquiring cardiovascular signals, these raw recordings alone are not sufficient to derive BP values. They must first be processed to extract physiologically meaningful features. Section IV, therefore, transitions from signal acquisition to signal processing and estimation techniques, examining how deterministic models, stochastic filters, and AI approaches transform raw signals into reliable BP estimates. Commonly used temporal features include pulse arrival time (PAT), defined as the interval between the R-wave of the ECG and the arrival of the pressure wave at a peripheral site [149], [150], [151], and PTT, the propagation time of the arterial pulse wave between two locations (e.g., from the heart to the fingertip). Both parameters are strongly correlated with arterial stiffness and BP and therefore provide the foundation for many estimation models introduced in this section.

Signal processing approaches in this field can be conceptually divided into three categories: 1) deterministic methods, which exploit analytical relationships such as the Moens–Korteweg equation or Windkessel modeling to link PTT or PAT to BP; 2) stochastic approaches, e.g., Wiener or Kalman filtering; and 3) data-driven methods, encompassing ML and DL algorithms, which have gained prominence for their ability to capture nonlinear relationships, integrate multimodal signals, and improve robustness against motion artifacts. This section reviews these categories as applied to noninvasive BP estimation, highlighting both their methodological principles and reported performance.

A. Deterministic Approaches

Deterministic methods estimate BP by relying on predefined mathematical relationships between measurable cardiovascular parameters and arterial pressure. These approaches do not require large training datasets but instead apply analytical models or direct feature extraction from physiological signals. The most common example is the use of PTT or PAT, both of which are inversely related to arterial stiffness and thus to BP.

A frequently applied model is the Moens–Korteweg equation, which links PWV to arterial elasticity [68], [106]. Since PWV is directly related to PTT, BP can be inferred through regression or calibration curves. Similarly, the Windkessel model represents the arterial system as a lumped electrical circuit, where BP can be estimated from changes in vascular compliance and resistance [105], [111]. These models provide interpreted and physiologically grounded relationships but often require calibration against reference measurements to ensure accuracy across individuals.

Beyond these methods, regression models have also been investigated; for example, a recent study applied multivariate linear regression to pulsed-wave Doppler ultrasound waveform indices in the carotid artery, demonstrating the feasibility of estimating local PWV as a pressure-independent surrogate of arterial stiffness [112]. Other approaches rely on threshold- or morphology-based analysis of oscillometric or PPG signals, such as identifying fiducial points on PPG waveforms, analyzing pulsewidth or amplitude, or applying curve-fitting methods to oscillometric envelopes [20]. While straightforward and computationally efficient, these techniques remain sensitive to signal noise, motion artifacts, and interindividual variability in vascular properties. A smaller body of work has explored frequency-domain representations of cardiovascular signals; for instance, Naufal et al. [23] applied spectrogram-based analysis of Korotkoff sounds for oscillometric estimation. Although such techniques are common in broader biomedical signal processing, their application to BP estimation has been relatively limited in the reviewed literature.

B. Stochastic Approaches

Stochastic approaches are methods that model signals and their underlying dynamics as random processes, aiming to estimate hidden states optimally in the presence of noise and uncertainty. Classical representatives include the Wiener filter, designed for stationary signals, and the Kalman filter, which recursively estimates time-varying states based on a combination of system dynamics and probabilistic noise models.

Although these filters are widely used in biomedical signal processing, their application to BP estimation is uncommon. A recent study employed an adaptive Kalman filtering scheme combined with a custom 4-LED PPG sensor to improve robustness against motion artifacts and enable continuous, noninvasive BP monitoring [152]. The Kalman filter operates by predicting the expected state of the signal (e.g., BP dynamics), updating this prediction with new measurements, and iteratively minimizing the estimation error, which makes it particularly suitable for noisy, real-time physiological data.

Overall, stochastic filtering offers a principled way to handle noise and uncertainty in physiological signals, but its adoption

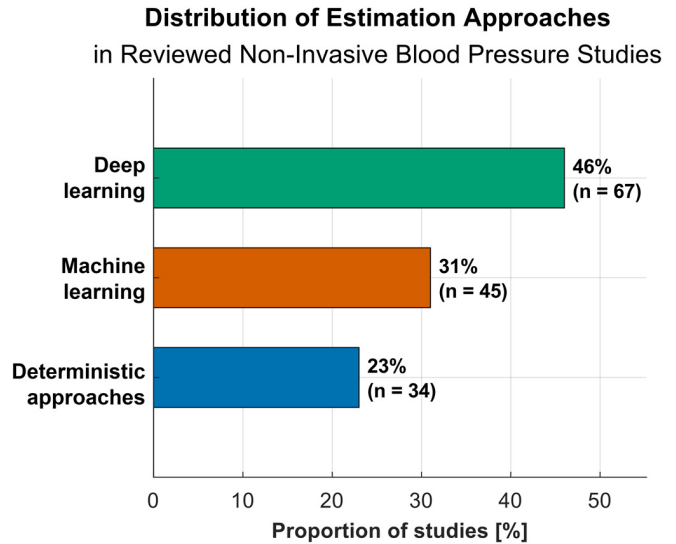


Fig. 9. Distribution of methods of estimating BP in the review.

in BP monitoring remains limited compared to deterministic or DL-based methods. Instead, some studies used probabilistic ML models, such as Bayesian model averaging (BMA) or Bayesian hierarchical models, which were therefore included under the ML category (Section IV-C1).

C. Data-Driven Approaches

Data-driven approaches, often grouped under the umbrella of AI, have become increasingly prominent in noninvasive BP estimation. Unlike deterministic models, which rely on predefined analytical relationships, AI techniques learn complex patterns directly from data. As shown in Fig. 9, most studies included in our work use either ML or DL methods, with only a quarter employing deterministic relationships. Within this category, a fundamental distinction is made between ML algorithms and DL architecture. ML typically operates on handcrafted or extracted features such as PTT, PAT, waveform morphology, or demographic covariates, whereas DL methods are capable of automatically learning hierarchical feature representations from raw signals or images.

A key advantage of data-driven approaches is their ability to cope with physiological variability, motion artifacts, and multimodal signal integration (e.g., combining PPG, ECG, SCG, or radar). However, their performance strongly depends on the availability and quality of training data, and overfitting remains a significant challenge. Moreover, many DL models lack interpretability, limiting their direct clinical adoption. This section, therefore, provides a structured taxonomy of ML (Section IV-C1) and DL (Section IV-C2) methods applied in the reviewed literature, highlighting their methodological principles, application to BP estimation, and reported performance.

Features are key attributes or characteristics used as inputs for ML and DL models. These attributes play a crucial role in optimizing model weights and influencing overall prediction performance and accuracy. Features can be of various types, including numerical, categorical, binary, or temporal, and their selection and processing are critical steps in model

MACHINE LEARNING WORKFLOW

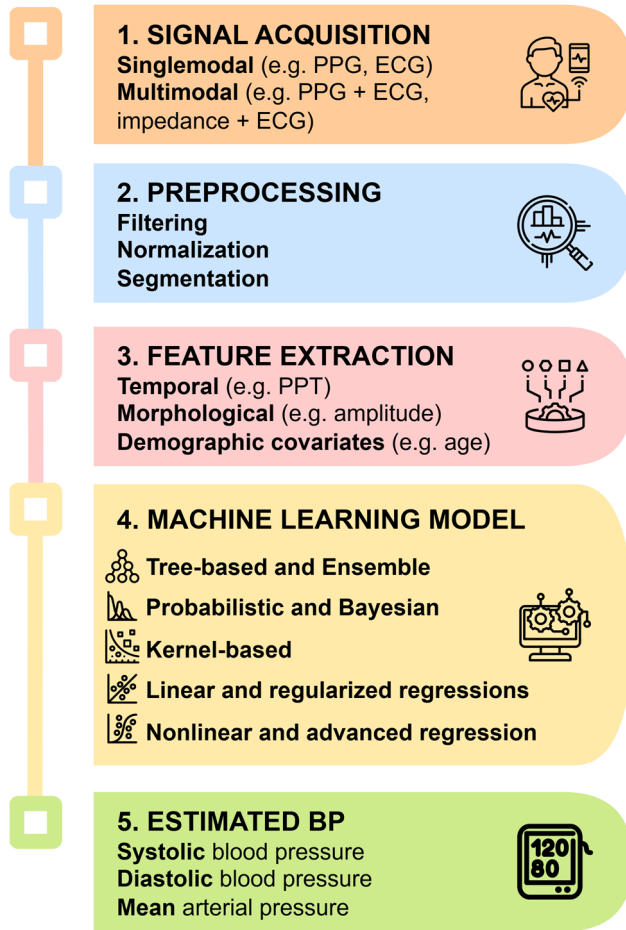


Fig. 10. General workflow of ML-based BP estimation.

development. In ML contexts, features are often extracted from raw data using techniques such as normalization, scaling, encoding categorical variables, or creating complex features by combining existing attributes. For example, in image processing, features may be represented by pixel intensities, textures, or histograms of oriented gradients. The research studied uses PTT or PAT signals as features along with individually measured signals or rPPG images. Features may include parameters from signals, for instance, total PTT time, PTT amplitude or ECG amplitude, R-wave amplitude, and R-wave duration. Additionally, the features can include patient's parameters such as body mass index (BMI) categories, age, gender, and diseases [153]. Some studies use fewer features (e.g., 5), while others use many features (up to around 42 [116]).

Instead of manual determination, algorithms such as ReliefF are utilized for feature creation. ReliefF is a feature selection algorithm that assesses the significance of individual features based on the differences between neighboring instances with different classes. [154].

1) *ML Approaches*: ML, a subdomain of AI, comprises algorithms that learn predictive mappings from empirical data. In the context of BP estimation, ML approaches have been

applied almost exclusively as supervised learning models, where physiological signal features are mapped to reference BP values. The general workflow of ML approaches for BP estimation is illustrated in Fig. 10. It consists of four main stages: signal acquisition, preprocessing, feature extraction, and model-based estimation.

The ML models used for BP estimation in the reviewed studies can be grouped into the following categories: 1) tree-based and ensemble models; 2) kernel-based model; 3) probabilistic and Bayesian model; 4) linear and regularized regression models; and 5) advanced and nonparametric regression models.

a) *Tree-Based and ensemble models*: Tree-based models are among the most applied due to their robustness and ability to handle complex feature interactions. Decision trees offer basic interpretability and serve as a foundation for more complex ensembles [53], [155]. RFs, which aggregate multiple trees to reduce variance, are widely used for BP estimation from PPG, ECG, or multimodal features [42], [54], [155]. Boosting-based algorithms such as adaptive boosting (AdaBoost) [45], [114] and category boosting (CatBoost) [150] have demonstrated improved accuracy in handling noisy and high-dimensional physiological data.

b) *Kernel-based models*: Support vector machines (SVMs) and vector support regression (SVR) have been extensively utilized in cuffless BP estimation. These models can capture nonlinear relationships between features and BP values using kernel functions. SVR is particularly suited for small-to-medium-sized datasets and has been applied to PPG and ECG-derived features such as PTT [37], [143].

c) *Probabilistic and bayesian models*: Bayesian methods enable uncertainty quantification and model combination. BMA [72] integrates predictions from multiple models weighted by their posterior probabilities. Hierarchical Bayesian models with single-point calibration have been proposed to personalize BP estimation by adapting model parameters to individual users [88].

d) *Linear and regularized regression models*: Linear regression and its multivariable variants have been used to model the direct relationship between physiological features and BP [85], [105]. To address multicollinearity, ridge regression introduces L2 regularization [61]. Despite their computational efficiency, these models struggle to capture the intricate, nonlinear cardiovascular dynamics that are crucial for BP estimation.

e) *Advanced and nonparametric regression models*: Nonlinear regression [55], nonparametric regression [63], and stepwise regression [100] provide greater modeling flexibility in scenarios where the functional relationship between features and BP is unknown or varies across individuals. Multi-instance regression (MIR) has been applied in settings involving time-segmented or multisensor data, enabling more granular BP estimation [108].

2) *DL Approaches*: DL is a subclass of ML techniques based on artificial NNs with multiple processing layers, collectively referred to as DNNs. These architectures are capable of modeling highly complex, nonlinear relationships between inputs and outputs. The general workflow of DL-based methods is shown in Fig. 11. Unlike ML approaches, DL models

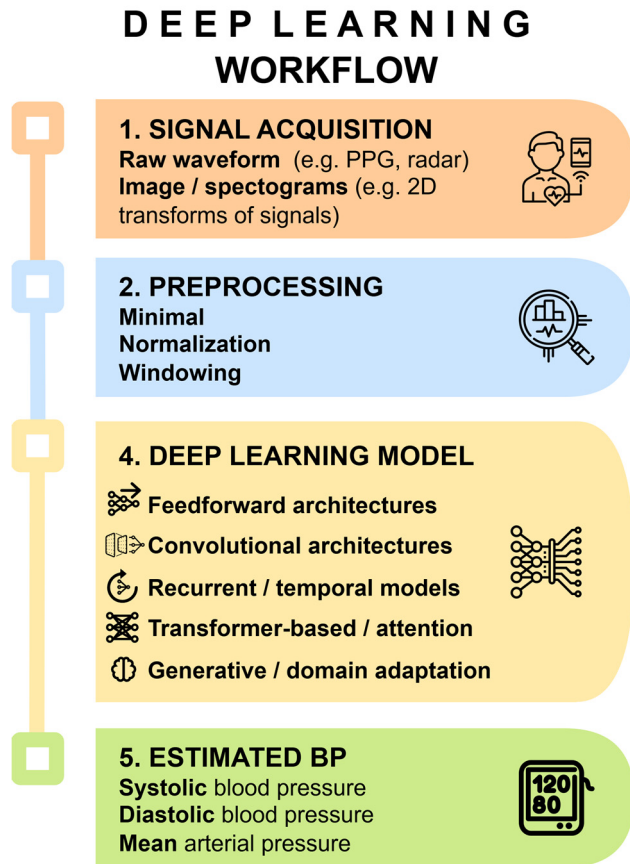


Fig. 11. General workflow of DL-based approaches for noninvasive BP estimation.

directly process raw waveforms or spectrograms with minimal preprocessing, automatically learning hierarchical features for BP estimation.

DL models used in BP estimation can be categorized based on their architectural family and functional characteristics, including feedforward, convolutional, recurrent, transformer, and generative models.

a) Feedforward architectures: The multilayer perceptron (MLP) represents the basic form of DNN, composed of fully connected layers with nonlinear activation functions. MLPs have been applied to regression tasks involving engineered features extracted from PPG, ECG, or SCG signals [156].

Dendritic neural regression (DNR) extends the MLP architecture by incorporating biologically inspired components such as synaptic, dendritic, membrane, and soma layers. This design improves the modeling of spatiotemporal dynamics and nonlinear input–output relationships, making it suitable for BP estimation under variable signal conditions [89].

b) Convolutional architectures: CNNs are widely used for processing structured inputs such as images and can also be applied to time-series data to extract local temporal features. However, for modeling long-range temporal dependencies, recurrent or attention-based architectures are often preferred. In BP estimation, CNNs are typically applied to PPG signals reshaped as 2-D images or spectrograms, allowing automatic extraction of morphological features relevant to systolic and diastolic pressure [69], [157].

The U-Net architecture, originally developed for biomedical image segmentation, has also been adapted for the precise localization of pulse wave features within PPG signals [158]. Its encode–decoder structure with skip connections enables multiscale feature learning, beneficial in distinguishing signal components associated with cardiovascular dynamics.

Siamese networks use two or more identical subnetworks to learn similarity between input pairs. In BP-related applications, Siamese models have been explored for subject-specific feature learning or to match signal patterns across sessions [60].

c) Recurrent and temporal models: Recurrent NNs (RNNs) are designed for sequential data processing. They maintain internal memory by feeding the previous time step’s hidden state into the next, which enables the modeling of temporal dependencies. In BP estimation, RNNs are used to learn dynamics from time-series physiological signals [21], [90].

LSTM networks improve upon standard RNNs by introducing gated memory cells that mitigate the vanishing gradient problem and capture long-term dependencies. LSTMs, or their subtype bidirectional LSTM (BLSTM), have demonstrated effectiveness in continuous BP monitoring tasks using raw or preprocessed PPG and ECG signals [136], [145], [159].

Gated recurrent units (GRUs) are simplified variants of LSTMs, using fewer parameters while retaining comparable temporal modeling capabilities. GRUs have been shown to be computationally efficient while maintaining performance in BP estimation applications [46], [50], [90].

Nonlinear autoregressive neural networks with exogenous inputs (NARX NNs) represent a class of recurrent models that incorporate external inputs alongside autoregressive signal history. These have been utilized for BP prediction based on combined signal history and contextual variables [84].

d) Transformer-based and attention models: KD-Informer integrates the Transformer framework with SE-ResNet modules and knowledge distillation for efficient PPG signal encoding. This architecture has been applied in BP estimation tasks requiring attention to long-range dependencies and computational efficiency [160].

e) Generative and domain adaptation models: CycleGAN is a generative adversarial network (GAN) variant designed for unpaired image-to-image translation. In BP-related research, it has been explored for domain adaptation between different sensing modalities or signal conditions [51].

DL models in BP estimation often operate on raw signals or their transformed representations (e.g., wavelet transforms and spectrograms). Compared to traditional ML approaches, DL eliminates the need for manual feature engineering by learning hierarchical feature representations directly from data. However, DL models generally require large datasets for training, are computationally demanding, and often act as black-box systems lacking interpretability—posing challenges in clinical adoption.

To improve generalization, recent studies have explored hybrid models combining CNNs or LSTMs with attention mechanisms or physiological priors. Domain adaptation techniques, such as adversarial training and transfer learning, are

TABLE I
PERFORMANCE OF CUFF-BASED METHODS AND THEIR ACCURACY

First Author	Signal	Technique	N. patients (n)	Error			Deviation			Z-score		Quality	
				Type	SBP (mmHg)	DBP (mmHg)	Type	SBP (mmHg)	DBP (mmHg)	SBP (-)	DBP (-)	Category	Score (-)
Alvarez Vazquez, Enrique [20]	AUS	Microphone and thresholds	20	-	-	-	STD	4.71	4.90	-	-	Medium	6
Argha, Ahmadrza [21]	AUS	BLSTM-RNN	155	MAE	1.70	3.40	SDE	3.70	5.00	0.46	0.68	Medium	6
Argha, Ahmadrza [30]	OSC	LSTM	350	ME	-1.20	1.80	$\pm$	5.90	8.80	-0.20	0.20	Very high	10*
Choi, Minh [31]	OSC	CNN	64	MAE	3.12	3.98	SDE	4.57	5.27	0.68	0.76	High	7
Hampson, Rory [28]	OSC	Deterministic algorithms	20	MAE	0.61	0.38	STD	7.14	5.91	0.09	0.06	High	7
Karimian, Sahar [25]	OSC	Best curve fitting cubic	80	Error	2.40	3.53	STD	4.25	7.12	0.56	0.50	High	8
Naufal, Dziban [23]	AUS	Deterministic algorithms	NS	Average error	3.46	3.10	STD	3.03	3.23	1.14	0.96	High	7
Pratiwi, Nurdina Gita [19]	AUS	Microphone and thresholds	85	ME	4.50	4.30	STD	5.70	5.50	0.79	0.78	Very high	9

* Fully complied with AAMI/ESH/ISO validation protocols.

Note: Rows highlighted in green (or bolded in print) indicate studies classified as very high quality. Rows highlighted in blue (or italicized in print) denote studies with the lowest z-score within each method group, excluding those already classified as very high quality.

also under investigation to enhance robustness across subjects, devices, and conditions.

V. EVALUATION OF TECHNIQUES

In this section, we compare how each study did with BP estimates. Table I provides an overview of cuff-based methods like oscillometric (OSC) and auscultatory (AUS). The overview of the cuffless methods is divided into two tables. Cuffless methods for measuring BP were intentionally divided into two subcategories: cuffless (experimental) and cuffless (database derived). This distinction was made to reflect the fundamental methodological differences between studies based on original signal acquisition in controlled or real-world environments and those relying on prerecorded datasets such as Multiparameter Intelligent Monitoring in Intensive Care (MIMIC), UCI, or PPG-BP. Experimental studies often involve direct interaction with measurement hardware, subject-specific calibration, and real-time signal variability, whereas database-based studies typically benefit from large-scale, standardized data but may lack contextual information such as sensor placement, environmental conditions, or subject behavior. By maintaining this separation, this review enables a more nuanced comparison of performance metrics and highlights the impact of data origin on model generalizability and clinical applicability. Therefore, Table II gives an overview of the cuffless experimental studies (including information on whether calibration was required—YES, NO, or NS if not specified), and Table III gives an overview of the cuffless database-based studies. Table IV provides an overview of contactless methods.

Most often, the new measurement methods were compared against cuff-based methods such as auscultation, where an experienced physician was present, or medical equipment based on the oscillometric principle, most often from OMRON, was used. Furthermore, it was also compared against Peña's method, i.e., volume-clamp methods, because of its continuous measurement against which they could compare. Within the databases, values were often compared against the measured invasive arterial BP (IABP).

A. Error Metrics Used for Evaluation

For consistency, only one accuracy metric was included per study. Mean absolute error (MAE) was prioritized, as it was the most frequently reported metric and provides a straightforward interpretation of average error magnitude.

MAE is also recommended by major validation standards for noninvasive BP measurement (AAMI/ESH/ISO standards [161]). When MAE was unavailable, mean error (ME) was used as the secondary metric due to its similar interpretability and frequent reporting. If neither MAE nor ME was provided, other measures such as mean difference (MD), mean absolute difference (MAD), correlation coefficient (R^2), or percentage error were adopted. In a few studies, the specific error definition was unclear. If the value is only in the SBP column, the authors did not report the difference between the systolic and DBP estimates and evaluated them together.

We also considered the deviations; where they were calculated, it was most often the standard deviation (STD) or sometimes the STD of the mean (SDE). In some cases, the deviation was not given only as $\pm$.

B. Standardization of Results Using Z-Scores

To enable a fair comparison across studies, standardized Z-scores were calculated on a study-wise basis, i.e., individually for each included study. For a given study i , the Z-score was computed as (1), where error_i represents the reported error metric (e.g., MAE and ME), and STD_i denotes the corresponding STD reported in the same study. This local standardization avoids the bias that would result from applying a global reference value and ensures that each Z-score reflects the relative accuracy within the specific measurement conditions and variability of the respective study

$$z_i = \frac{\text{error}_i}{\text{STD}_i}. \quad (1)$$

Only data points with complete information were considered in the analysis—specifically, those studies that reported both an error value and its corresponding STD. Consequently, studies presenting results solely as percentage errors, correlation coefficients, or other nonstandard metrics were excluded from the Z-score transformation.

C. Public Databases for Cuffless BP Estimation

In cases where studies focused solely on estimating BP from signals, particularly from PPG, publicly available databases were often utilized:

MIMIC Database provides comprehensive intensive care unit (ICU) patient data, including physiological signals, laboratory results, and clinical notes, facilitating research in

TABLE II
PERFORMANCE OF CUFFLESS BP METHODS

First Author	Signal	Technique	N, patients (n)	Error			Deviation			Z-score		Quality		Calibration
				Type	SBP (mmHg)	DBP (mmHg)	Type	SBP (mmHg)	DBP (mmHg)	SBP (-)	DBP (-)	Category	Score (-)	
Al-Hashem, Munder [80]	PPG, ECG	Polynomial surface fitting	17	MAE	4.13	6.89	-	-	-	-	-	Medium	6	NO
Baek, Sanghyun [95]	Green light PPG	CNN	26	MAE	5.28	4.92	STD	1.80	2.42	2.93	2.03	Medium	6	NO
Carek, Andrew M. [103]	PPG, BCG	Deterministic algorithms	16	Error	5.60	2.60	±	2.70	1.20	2.07	2.17	Medium	6	NS
Chakraborty, Ayan [63]	PPG	Nonparametric Regression	162	MAE	2.50	2.12	SDE	2.07	1.79	1.21	1.18	High	8	NS
Chakraborty, Ayan [64]	PPG	Deterministic algorithms	31	MAE	3.63	2.28	STD	3.32	2.07	1.09	1.10	High	7	YES
Chen, Kexin [77]	PPG, ECG	Optimization algorithms	98	ME	-0.39	-0.11	STD	6.54	3.61	-0.06	-0.03	High	8	YES
Das, Mousumi [96]	PPG, ECG, SCG	Left ventricular ejection time (LVET)	10	MAE	2.59	3.17	STD	2.58	3.33	1.00	0.95	Medium	6	YES
Dave, Tejal [37]	PPG, ECG	Support Vector Regression	87	ME	0.04	1.99	STD	4.75	4.78	0.01	0.42	High	8	NS
De Marchi, Beatrice [85]	PPG, ECG	Linear regression model	NS	MAE	>6	>6	-	-	-	-	-	High	7	YES
Diehl, Matthias [117]	Multimodal sensors	1D-CNN	27	ME	-1.03	4.63	STD	11.81	11.35	-0.09	0.41	High	7	YES
Gogiberidze, Nana [73]	PPG, ECG	CardioQVARK	167	MAD	3.44	3.21	±	2.50	2.82	1.38	1.14	Very high	10*	NS
Guo, Cheng-Yan [106]	piezoelectric sensor	Moens-Korteweg equation	32	Mean Difference	-0.63	-1.14	STD	5.14	4.08	-0.12	-0.28	High	7	NO
He, Shan [88]	PPG, ECG	Hierarchical model with a single calibration measurement	10	MAD	4.36	-	STD	1.17	-	3.73	-	Medium	6	YES
Jana, Biswabandhu [111]	CW Doppler	Windkessel BP model with SVR	85	ME	-0.26	-0.08	SDE	9.10	7.85	-0.03	-0.01	Very high	9	NS
Jinu, Agnes [105]	piezosensor	Multivariable linear regression model	NS	MAE	1.33	1.14	-	-	-	-	-	Medium	5	NS
Khan Mamun [154]	PPG, ECG	ReliefF, Multivariate symmetric uncertainty	NS	MAE	4.27	5.01	-	-	-	-	-	Medium	6	NS
Koay, Yi Yun [68]	PPG, ECG	Moens-Korteweg equation	30	Percentage error (%)	7.01	6.50	-	-	-	-	-	Medium	6	YES
Kumar, N. Vajresh [118]	Fiber Bragg Grating (FBG) sensor	Deterministic algorithms	35	Average accuracy (%)	90.02	93.42	STD	2.13	2.57	-	-	High	7	YES
Landry, Cederick [86]	PPG, ECG**	NN	5	MAE	5.85	-	±	0.73	-	8.01	-	Medium	6	YES
Liu, Zeng-Ding [45]	PPG, ECG, PPW	AdaBoost	3,077	MAE	7.38	5.30	SDE	9.57	6.52	0.77	0.81	Very high	9	YES
Louka, Kyriolos [71]	PPG, ECG	Deterministic algorithms	20	R (-)	0.36	0.04	p	0.13	0.37	2.77	0.11	Medium	5	YES
Mahmuda [67]	PPG	DNN	65	R ² (-)	0.96	0.95	-	-	-	-	-	Medium	6	NS
Mansourinezhad, Paria [39]	PPG**	Deterministic algorithms	120	MAE	6.63	3.96	STD	5.59	3.82	1.19	1.04	High	8	NS
Marzoorati, Davide [79]	PPG, PCG	Deterministic algorithms	20	MAD	1.83	3.06	STD	3.76	7.55	0.49	0.41	Medium	6	YES
Miao, Fen [108]	ECG, PPW	Multi-instance regression model	85	MAE	6.13	4.54	STD	7.76	5.52	0.79	0.82	Very high	9	YES
Mou, Hanlin [46]	PPG	Transfer learning - CNN-GRU	279 signals	MAE	6.08	4.20	-	-	-	-	-	Medium	6	NS
Nandi, Pratyush [70]	PPG, ECG	CNN-LSTM	30	MAE	2.57	3.44	STD	2.39	2.63	1.08	1.31	High	8	NS
Nguyen, Duc Huy [50]	PPG	CNN + GRU with Quality check model	154	MAE	3.71	2.36	SDE	6.23	3.93	0.60	0.60	High	8	NS
Osman, Deen [114]	bioimpedance sensor	AdaBoost regression model	5	MAE	4.38	3.63	STD	2.89	2.42	1.52	1.50	Medium	6	NS
Pediaditis, M. [42]	IR PPG	Random Forest regression	11	MAE	3.15	3.53	-	-	-	-	-	Medium	6	NO
Scardulla, Francesco [44]	PPG with tightening system	Deterministic algorithms	12	Mean Difference	4.87	5.71	STD	6.06	7.56	0.80	0.76	High	7	NS
Shen, Zhan [72]	PPG, ECG	Bayesian Model Average	30	MAE	2.13	1.42	±	5.68	5.11	0.38	0.28	High	7	YES
Shi, Wei [152]	PPG	Adaptive Kalman Filtering	10	MAE	2.64	2.87	STD	2.34	2.69	1.13	1.07	Medium	6	YES
Skoric, James [99]	VCG	CNN	62	MAE	3.40	2.20	±	4.80	3.00	0.71	0.73	High	7	YES
Song, Kwangsub [81]	PPG, ECG	DNN	110	MAE	4.20	4.20	SDAE	3.20	3.00	1.31	1.40	Very high	9	YES
Tabei, Fatemehsadat [92]	PPG	Linear regression model	6	MAE	2.07	2.12	STD	2.06	1.85	1.00	1.15	Medium	6	NS
Tseng, Chao-Hsiung [163]	microwave NFSIL	Deterministic algorithms	10	MD	0.26	0.56	STD	3.67	6.99	0.07	0.08	Medium	6	YES
Vajananar, Tipnirin [149]	PPG, ECG	Deterministic algorithms	4	MAD	4.69	10.19	-	-	-	-	-	Medium	6	YES
Wang, Ting-Wei [115]	IPG	Deterministic algorithms	30	MAE	2.01	2.26	±	1.40	1.43	1.44	1.58	High	7	YES
Xu, Lirui [113]	Ultrasound PP	Deterministic algorithms	7	MAE	4.60	3.70	±	5.90	4.80	0.78	0.77	Medium	6	NO
Yang, Bufang [116]	BIOZ, ECG	Random forest	13	MAE	2.17	1.71	±	0.18	0.18	12.06	9.50	High	7	YES
Yang, Sen [83]	PPG, ECG	Hybrid CNN model with LSTM	45	MAE	4.43	3.23	STD	6.09	4.75	0.73	0.68	High	7	NO
Yao, Pan [55]	PPG	Nonlinear regression	33	MAE	3.23	2.73	SDE	4.47	3.61	0.72	0.76	High	8	NO
Zhang, Qian [107]	Pressure sensor	Back Propagation Neural Network	85	MAE	2.16	1.96	-	-	-	-	-	High	8	NO
Zhang, Yandong [100]	PPG, ECG, BCG	Stepwise regression	20	MAE	6.84	5.46	-	-	-	-	-	Medium	6	YES
Zhang, Yuxuan [94]	PPG	NN	34	MAE	4.44	3.68	STD	5.68	3.36	0.78	1.10	High	7	NO

* Fully complied with AAMI/ESH/ISO validation protocols.

** Investigated dynamic measurements.

Note: Rows highlighted in **green** (or bolded in print) indicate studies classified as very high quality. Rows highlighted in **blue** (or italicized in print) denote studies with the lowest z-score within each method group, excluding those already classified as very high quality.

predictive modeling and patient monitoring. The main versions include MIMIC, MIMIC II and MIMIC III, with each newer version extending and improving on the previous one [48], [160].

Mindray dataset is a private dataset from Shenzhen Mindray Biomedical Electronics Company Ltd. Data were collected using a multiparameter physiological monitor in the ICU. The PPG signal was obtained via a transmissive finger cuff device, while the invasive pressure signal was obtained from a brachial artery cannula [48], [160].

PPG-BP Database is a public dataset that was collected from 219 adult subjects and currently contains 657 PPG waveform segments. The arterial BP (ABP) is measured using the Omron HEM-7201 (Omron Company, Kyoto, Japan) upper arm BP monitor [52].

UCI ML Repository is a subset of the MIMIC II waveform database. The UCI dataset contains 12 000 segments,

with each segment containing continuous PPG, ECG, and invasively measured ABP signals, with a sampling rate of 125 Hz [61].

Queensland Database from The University of Queensland Vital Signs Dataset is a public database that consists of vital sign parameters, including PPG and the corresponding BPs, extracted from 32 patients during anesthesia [62].

Kangdollar Database is a public dataset consisting of two parts of information, including the standard diastolic pressure, standard systolic pressure, physical characteristic parameters of the subjects, and preprocessed PPG and ECG waveforms [54], [78].

These datasets have facilitated reproducibility and large-scale training of AI models, but they may lack contextual information such as sensor placement, environmental conditions, or subject behavior, which limits their direct clinical applicability.

TABLE III
PERFORMANCE OF CUFFLESS BP METHODS USING DATABASE-DERIVED DATA

First Author	Signal	Technique	Database	N. patients (n)	Error			Deviation			Z-score		Quality		Calibration
					Type	SBP (mmHg)	DBP (mmHg)	Type	SBP (mmHg)	DBP (mmHg)	SBP (-)	DBP (-)	Category	Score (-)	
Aldein, Elham Alaa [158]	ECG	U-NET + Random Forest	MIMIC III	NS	MAE	6.00	2.50	±	4.49	3.70	1.34	0.68	Low	4	NO
An Dong, Bui [56]	PPG	CNN	MIMIC II	NS	MAE	4.19	2.25	STD	5.91	3.21	0.71	0.70	Low	4	NO
Chakraborty, Ayan [64]	PPG	Deterministic algorithms	MIMIC III	150	MAE	3.63	2.28	STD	3.32	2.07	1.09	1.10	Medium	6	NS
Chen, Qinda [54]	PPG	CNN-LSTM-LightGBM combination model	Kangdollar dataset	1,200	MAE	2.39	2.62	-	-	-	-	-	Medium	6	NS
Chen, Xiaohui [78]	PPG, ECG	Random forest	Kangdollar dataset	1,060	MAE	4.45	3.95	-	-	-	-	-	Medium	6	NS
Chu, Yan [162]	PPG	Transformer-based deep learning architecture	MIMIC III	1,732	MAE	2.52	1.37	STD	2.43	1.89	1.04	0.72	Medium	6	NS
Dal Pont, Mauricio P. [164]	PPG	Deterministic algorithms	MIMIC II	34	ME	-2.37	-0.4	STD	6.86	7.47	-0.35	-0.05	Medium	5	YES
El Hajj, Chadi [159]	PPG	LSMT	MIMIC II	NS	MAE	3.23	1.59	STD	4.76	1.96	0.68	0.81	Low	3	NS
El Hajj, Chadi [90]	PPG, ECG	RNN-GRU	MIMIC II	942	MAE	5.77	3.33	STD	8.52	5.02	0.68	0.66	Medium	5	NS
Ghosh, Aayushman [150]	PPG	CatBoost model	MIMIC II	90	MAE	10.07	5.02	-	-	-	-	-	Medium	6	NS
Gupta, Shresth [49]	PPG	CNN-LSTM	MIMIC II	220	MAE	4.19	2.36	STD	6.66	3.42	0.63	0.69	Medium	6	NS
Ji, Junkai [89]	PPG, ECG	DNR	MIMIC	≈100	MAE	2.16	1.16	STD	1.01	0.46	2.14	2.52	Medium	6	NS
Khalid, Syed Ghufraan [62]	PPG	Two-step algorithm using ML	Queensland and MIMIC II	282	Mean Difference	0.07	0.08	STD	7.10	6.00	0.01	0.01	Medium	6	NS
Khan Mamun [154]	PPG, ECG	Multivariate symmetric uncertainty, ReliefF	MIMIC II	3,295	MAE	4.27	5.01	-	-	-	-	-	Medium	6	NS
Kim, Minseong [57]	PPG	DL + initial blood pressure value	MIMIC III	4,200	MAE	1.90	-	STD	2.93	-	0.65	-	Medium	6	YES
Ko, Pin-You [69]	PPG, ECG	CNN	MIMIC III	300	MAE	3.63	2.50	STD	11.00	10.00	0.33	0.25	Medium	6	NS
Li, Shichao [47]	PPG	TAS model of NN	MIMIC II / volunteers	200 / 20	MAE	2.85	1.48	STD	4.65	2.31	0.61	0.64	Medium	6	NS
Liu, Xinhua [59]	PPG	CNN (PPGLAB)	MIMIC III	942	ME	-2.62	1.36	STD	7.78	5.74	-0.34	0.24	Medium	6	NS
Ma, Chenbin [160]	PPG	KD-Informer (Transfer learning)	MIMIC	708	ME	0.02	0.01	STD	5.93	3.87	0.00	0.00	Medium	6	NS
Ma, Chenbin [48]	PPG	SMART-BP	Midray / MIMIC	683	MAE	2.52	1.47	STD	3.26	2.22	0.77	0.66	Medium	6	NO
Maharajan, Ramakrishnan [165]	PPG	Deterministic algorithms	MIMIC II	25	ME	0.40	-	±	4.85	-	0.08	-	Low	4	YES
Mehrabadi, Milad Asgari [51]	PPG	CycleGAN	MIMIC II	92	MAE	2.89	3.22	STD	4.52	4.67	0.64	0.69	Medium	6	NS
Paviglianiti, Annunziata [87]	PPG, ECG	ABPNET	MIMIC	37	RMSE	3.60	3.20	-	-	-	-	-	Medium	5	NS
Samimi, Hamed [166]	PPG	ANN	MIMIC II	390	MAE	13.97	5.85	SDE	16.86	7.16	0.83	0.82	Medium	6	NO
Schlesinger, Oded [60]	PPG	CNN, SIAMESE network	MIMIC II	1,459	MAD	5.95	3.41	STD	6.69	3.97	0.89	0.86	Medium	6	YES
Senturk, Umit [84]	PPG, ECG	NARX-NN and LSTM-NN	MIMIC	NS	ME	0.02	0.04	STD	2.21	1.22	0.01	0.03	Medium	6	NS
Shoebi, Fatemeh [53]	PPG	Decision tree regression ML model	MIMIC II	101	MAE	2.10	1.40	STD	5.09	3.79	0.41	0.37	Medium	6	NS
Tjahjedi, Hendrana [52]	PPG	BLSTM with time-frequency analysis	PPG-BP	219	F1 (%)	97.39	-	-	-	-	-	-	Medium	6	NO
Wang, Weinan [61]	PPG	Ridge regression	UCI	348	MAE	6.17	8.46	STD	3.66	5.36	1.69	1.58	Medium	6	NO

Note: Rows highlighted in **green** (or bolded in print) indicate studies classified as very high quality. Rows highlighted in **blue** (or italicized in print) denote studies with the lowest z-score within each method group, excluding those already classified as very high quality.

D. Quality Assessment of Included Studies

To provide a structured comparison of the studies included, we applied a custom quality assessment framework. Each study was evaluated across five criteria: 1) clinical validation (prospective testing in patient populations); 2) sample size (≥ 85 participants as recommended by AAMI/ESH/ISO standards [161], 30–84, or < 30); 3) type of data (originally acquired versus database derived or simulated); 4) methodological transparency (clarity of signal acquisition, preprocessing, algorithm description, and evaluation procedure); and 5) compliance with validation protocols (formal adherence to AAMI/ESH/ISO [161] standards versus informal or no evaluation). Each criterion was scored on a scale of 0–2 points, resulting in a maximum of 10 points per study. Calibration requirements were additionally recorded as a separate attribute, given their relevance to cuffless PPG-based approaches.

Although early stage or unvalidated methods were included to capture emerging technologies, the quality assessment framework allows these studies to be clearly distinguished from clinically validated works. This approach preserves the breadth of technological coverage while maintaining analytical rigor by enabling readers to interpret results in the context of study quality.

Based on the aggregated score, studies were classified into five categories: **very high quality** (9–10 points), **high quality**

(7–8 points), **medium quality** (5–6 points), **low quality** (3–4 points), and **very low quality** (0–2 points). This extended classification allowed for finer differentiation between studies and provided a more nuanced interpretation of methodological rigor, clinical relevance, and reproducibility. Studies scoring 10 fully complied with international validation protocols such as AAMI/ESH/ISO [161]. Those scoring 9 typically either exceeded the permitted error margins or narrowly missed specific criteria (e.g., including slightly fewer participants than the minimum 85 required).

The number of patients involved in studies examining these methods ranged from over 1500 participants to fewer than 10. Measurement settings also varied widely, including hospitals, research laboratories, and academic environments. If the study did not specify the number of patients, not specified (NS) is indicated in the Tables I–IV.

E. Evaluation of Individual Studies

To provide a detailed assessment, individual studies were analyzed separately within four methodological categories corresponding to Tables I–IV. Each table compiles the key error metrics, standardized Z-scores, quality ratings, and calibration requirements (when available). In the following, we summarize the main findings for each category, highlighting representative studies and overall performance trends.

TABLE IV
PERFORMANCE OF CONTACTLESS BP METHODS

First Author	Signal	Technique	N. patients (n)	Error			Deviation			Z-score		Quality	
				Type	SBP (mmHg)	DBP (mmHg)	Type	SBP (mmHg)	DBP (mmHg)	SBP (-)	DBP (-)	Category	Score (-)
Geng, Fanglin [125]	mmWave radar	Radar-detected transit time (RDT)	27	Error	5.54	4.68	±	7.62	6.15	0.73	0.76	High	7
Goudarz, Reza Heydari [139]	rPPG	AdaBoost regression model	118	MAE	5.74	4.86	STD	3.66	3.27	1.57	1.49	Very high	9
Goudarz, Reza Heydari [140]	rPPG	Non-Linear Random Forest Regression	40	ME	0.45	-0.20	STD	12.39	6.41	0.04	-0.03	High	7
Huang, Yukai [141]	rPPG	SVR	41	MAE	4.78	5.52	STD	2.1	2.3	2.28	2.40	High	8
Iwashita, Yuki [143]	IR camera	SVM	1	RMSE	6.60	6.44	-	-	-	-	-	Medium	6
Lumyong, Chayanin [136]	rPPG	LSTM, CHROM	NS	MAE	10.04	8.37	-	-	-	-	-	Medium	5
Ochi, Hinako [121]	microwave reflection	Machine learning	9	RMSE	5.00	-	-	-	-	-	-	Medium	6
Ochi, Hinako [124]	microwave reflection	Random forest	NS	MAPE (%)	3.40	-	-	-	-	-	-	Low	4
Ochi, Hinako [129]	microwave reflection	Multiple regression analysis	5	MAE	3.49	-	STD	2.32	-	1.50	-	Medium	6
Parashiva, Praveen K. [145]	rPPG	LSTM	NS	RMSE	17.79	10.83	-	-	-	-	-	Medium	6
Qiao, Donghao [137]	rPPG	Deep learning	49	MAE	6.70	9.60	-	-	-	-	-	High	8
Ran, You [122]	mmWave radar	Encoder-Decoder NN + CNN	20	Deviation rate (%)	9.00	3.69	-	-	-	-	-	High	7
Shaikh, Mohd Rizwan [142]	IR camera	DNN	20	MAE	2.27	2.51	STD	3.11	3.44	0.73	0.73	High	7
Shi, Haotian [120]	radar	Random forest regression	25	MAE	7.20	6.30	STD	6.73	3.85	1.07	1.64	Medium	5
Subramani, Saikiran [147]	EMF	Deterministic algorithms	34	R (-)	0.93	0.77	-	-	-	-	-	Medium	5
Wang, Yong [126]	mmWave FMCW radar	TVE-EMD + DSFNN-BP	40	MAE	4.16	3.87	STD	5.19	5.11	0.80	0.76	High	7
Wu, Bing-Fei [132]	rPPG	CNN	1,138	MAE	10.59	5.95	STD	9.07	5.76	1.17	1.03	High	8
Wu, Bing-Fei [138]	rPPG	Feature- and Signal-Combined Network	1,143	MAE	11.54	8.09	STD	1.32	0.70	8.74	11.56	Very high	9
Wu, Bing-Jhang [131]	rPPG	CNN	93	MAE	6.48	5.06	STD	5.72	4.32	1.13	1.17	High	8
Wuerich, Carolin [130]	rPPG	DL, Separate MLP	5	MAE	5.50	3.73	STD	4.52	2.86	1.22	1.30	Medium	6
Zheng, Zhi [123]	dual radar	ANN	27	MAE	9.51	6.69	STD	12.13	8.12	0.78	0.82	Medium	6

Note: Rows highlighted in **green** (or bolded in print) indicate studies classified as very high quality. Rows highlighted in **blue** (or italicized in print) denote studies with the lowest z-score within each method group, excluding those already classified as very high quality.

1) *Cuff-Based Methods*: Table I focuses on cuff-based methods for measuring BP, including auscultatory and oscillometric techniques. For example, Pratiwi et al. [19] achieved relatively strong results (ME 4.50 and 4.30 mmHg for SBP and DBP; Z-scores 0.79 and 0.78) in a cohort of 85 participants, earning a *very high* quality rating (score 9). The study's adherence to validation protocols and adequate sample size bolsters the reliability of its findings, despite their MEs being marginally higher than certain AI-based methodologies. Similarly, Karimian et al. [25] demonstrated solid performance using cubic curve fitting (errors of 2.40 and 3.53 mmHg; Z-scores of 0.56 and 0.50) with 80 participants, corresponding to a *high* quality classification (score of 8). In contrast, Alvarez Vazquez et al. [20] relied on microphone-based auscultation in a small sample ($n = 20$) without standardized deviation reporting, leading to a medium quality rating (score 6) and limited reproducibility.

Generally, the results confirm that while deterministic methods can achieve competitive accuracy under controlled conditions, studies employing larger cohorts and advanced algorithms (e.g., DL) tend to yield lower Z-scores and higher quality classifications, underscoring their greater robustness and clinical relevance.

2) *Cuffless Experimental Methods*: Table II summarizes cuffless experimental methods, with calibration requirements emerging as a key differentiator across studies. Among calibration-free approaches, Dave et al. [37] achieved strong performance with support vector regression (ME 0.04 and 1.99 mmHg; Z-scores 0.01 and 0.42) in 87 subjects, classified as *high* quality (score 8). Similarly, Nguyen and Chao [50] reported balanced MAE values (3.71 and 2.36 mmHg) with Z-scores of 0.60, also without explicit calibration, demonstrating that AI-driven methods can maintain accuracy while avoiding the complexity of repeated reference measurements. By contrast, deterministic regression by Marzorati et al. [79] required calibration but still achieved competitive accuracy (MAD 1.83

and 3.06 mmHg; Z-scores 0.49 and 0.41) in a small cohort ($n = 20$, *medium* quality, score 6).

On the other hand, several studies relying on calibration highlighted the tradeoff between methodological complexity and accuracy. For instance, Chen et al. [77] combined PPG and ECG signals using optimization algorithms, obtaining near-zero MEs (-0.39 and -0.11 mmHg; Z-scores -0.06 and -0.03) with a *high* quality score (8). Similarly, Mansourinezhad et al. [39] demonstrated acceptable accuracy (MAE 6.63 and 3.96 mmHg; Z-scores 1.19 and 1.04) across 120 participants, also classified as *high* quality (8.5).

Taken together, these findings suggest that calibration-free AI models can already achieve robust accuracy and lower Z-scores, whereas deterministic or regression-based methods often depend on calibration to remain competitive.

3) *Cuffless Database-Driven Methods*: Table III summarizes cuffless methods derived from public and private databases such as MIMIC, UCI, or Kangdollar. Unlike experimental studies, most database-based approaches did not explicitly report calibration requirements, and often, this aspect remained unspecified. While several studies involved enormous sample sizes—for instance, Kim et al. [57] with 4200 subjects or Chu et al. [162] with 1732 participants—the overall quality ratings remained modest, typically in the *medium* (score 5–6) or even *low* category (score 3–4). This reflects the inherent limitation of secondary datasets; although they provide statistical power, researchers cannot control sensor placement, environmental conditions, or patient behavior, nor can they validate measurements in real time.

Despite these limitations, AI-driven methods demonstrated competitive accuracy. Ji et al. [89], for example, reported MAE of 2.16 and 1.16 mmHg (Z-scores 2.14 and 2.52) using multimodal ECG and PPG signals, whereas Ma et al. [160] achieved near-zero MEs (0.02 and 0.01 mmHg) across 708 patients with a transfer learning approach. Still, both studies were only classified as *medium* quality due to reliance

on database data. Similarly, Senturk et al. [84] combined LSTM and NARX NNs to obtain almost negligible errors (ME 0.02 and 0.04; Z-scores 0.01 and 0.03), yet the lack of reported patient details and dependence on MIMIC reduced the robustness of their evaluation.

Taken together, database-derived studies offer the advantage of large cohorts and allow for rapid benchmarking of AI methods, but their methodological quality is inherently capped by the absence of clinical control.

4) *Contactless Methods*: Table IV summarizes contactless methods, including radar, rPPG, and camera-based approaches. Overall, these studies exhibited the largest variability, both in terms of accuracy and methodological quality. Shaikh et al. [142] achieved the strongest performance using an IR camera combined with a DNN model (MAE 2.27 and 2.51 mmHg; Z-scores 0.73 and 0.73) in 20 participants, rated as *high quality* (score 7). Similarly, Goudarz et al. [139] reported promising results with rPPG (MAE 5.74 and 4.86 mmHg; Z-scores around 1.5) across 118 subjects, earning a *very high quality* classification (score 9). These examples demonstrate that contactless sensing can approach clinically relevant accuracy under controlled conditions.

However, many studies showed considerably higher errors or extreme variability. For instance, Wu et al. [138] analyzed more than 1100 participants with an advanced feature- and signal-combined network for rPPG but reported high MAE values (11.54 and 8.09 mmHg) alongside very large Z-scores (8.74 and 11.56). Likewise, radar-based methods such as Zheng et al. [123] yielded MAE of 9.51 and 6.69 mmHg with only *medium* quality classification (score 6), reflecting both technological immaturity and sensitivity to environmental conditions.

Contactless methods currently represent the least consistent category. Although isolated studies with IR imaging or carefully processed rPPG achieved high or very high quality ratings, the majority exhibited substantial variability and higher standardized errors.

F. Evaluation Across Study Categories

To compare the performance of the studies included, the standardized Z-scores described above were aggregated at the category level. The categories were defined consistently with the structure of Tables I–IV, namely, cuff-based methods, cuffless experimental methods, cuffless database-derived methods, and contactless methods. For each category, the mean Z-score $\bar{z}$ and the mean absolute Z-score $|\bar{z}|$ were then calculated.

To assess the statistical confidence of the mean Z-scores, 95% confidence intervals (CIs) were computed using the following equation:

$$CI = \bar{z} \pm t_{(N-1)} \cdot \frac{s}{\sqrt{N}} \quad (2)$$

where s is the STD of the Z-scores within the category, N is the number of included studies, and t_{N-1} is the critical value from the Student's t -distribution for a two-tailed 95% confidence level and $N - 1$ ° of freedom.

Only data points with complete information (i.e., both the error value and its corresponding STD) were included in the

TABLE V
MEAN Z-SCORES AND 95% CI FOR BP MEASUREMENT METHODS

Measurement Methods	Z-score (Mean)	Z-score (Mean)	95% CI	N
SBP	Cuff-based	0.50	(0.12 to 0.88)	7
	Cuffless (experimental)	1.54	(0.47 to 2.61)	34
	Cuffless (database-derived)	0.63	(0.38 to 0.89)	23
	Contactless	1.67	(0.40 to 2.94)	13
DBP	Cuff-based	0.56	(0.28 to 0.84)	7
	Cuffless (experimental)	1.11	(0.34 to 1.88)	32
	Cuffless (database-derived)	0.67	(0.41 to 0.91)	21
	Contactless	1.97	(0.10 to 3.84)	12

TABLE VI
MEAN Z-SCORES AND 95% CI BY STUDY CATEGORY

Study Category	Z-score (Mean)	Z-score (Mean)	95% CI	N
SBP	Low	0.70	(-0.01 to 1.41)	4
	Medium	1.17	(0.68 to 1.67)	36
	High	1.14	(0.30 to 1.98)	28
	Very high	1.68	(-0.29 to 3.65)	9
DBP	Low	0.73	(0.59 to 0.87)	3
	Medium	0.83	(0.59 to 1.06)	32
	High	1.08	(0.42 to 1.74)	28
	Very high	2.02	(-0.60 to 4.64)	9

analysis. The summary statistics of Z-scores for each method category are presented in Table V.

The comparison of the four method categories based on mean Z-scores and mean absolute Z-scores revealed clear differences in accuracy and variability (Table V). Cuff-based methods yielded the lowest mean absolute Z-scores (SBP 0.56; DBP 0.56) with the narrowest CIs that suggest highly consistent performance across studies. Cuffless database-derived methods performed similarly well (SBP 0.69; DBP 0.67), also exhibiting relatively tight interval widths. In contrast, cuffless experimental methods showed substantially higher values (SBP 1.55; DBP 1.13) and broader intervals, indicating increased variability among studies. Contactless approaches produced the largest mean absolute Z-scores (SBP 1.67; DBP 1.97) and the widest CIs, reflecting considerable heterogeneity and overall lower reliability.

G. Evaluation Across Studies by Quality

When stratifying studies by methodological quality, mean absolute Z-scores for both SBP and DBP showed distinct trends (Table VI). All values reported below refer to mean absolute Z-scores. *Low* quality studies yielded the lowest values (SBP: 0.70; DBP: 0.73), although these estimates were based on only four studies and cannot be considered robust. *Medium* quality studies achieved moderate values (SBP: 1.21; DBP: 0.83) with comparatively narrow CIs, indicating greater consistency. *High* quality studies showed a similar pattern (SBP: 1.16; DBP: 1.10), suggesting that these two quality levels provide the most reliable and reproducible performance. In contrast, *very high* quality studies reported the largest values (SBP: 1.73; DBP: 2.02), together with the widest CIs,

TABLE VII
MEAN Z-SCORES AND 95% CI BY CALIBRATION REQUIREMENT

Calibration Requirement	Z-score (Mean)	Z-score (Mean)	95% CI	N
SBP	YES	1.84	1.89 (0.46 to 3.22)	20
	NO	1.01	1.04 (0.52 to 1.51)	11
	NS	0.75	0.78 (0.49 to 1.01)	25
	NO (incl. NS)*	0.83	0.86 (0.60 to 1.06)	36
DBP	YES	1.18	1.19 (0.08 to 2.28)	17
	NO	0.86	0.91 (0.49 to 1.24)	11
	NS	0.80	0.80 (0.54 to 1.06)	25
	NO (incl. NS)*	0.82	0.84 (0.61 to 1.02)	36

* NS values were conservatively treated as NO (no calibration).

reflecting substantial heterogeneity likely attributable to more complex study designs or diverse populations.

The analysis demonstrates that *medium* and *high* quality studies achieved the most consistent accuracy, while *low* quality studies lacked robustness due to small sample sizes. Studies of *very high quality* showed the least consistent accuracy but the greatest heterogeneity of data.

H. Evaluation by Calibration Requirement

An additional stratification was performed according to calibration requirements (Table VII). Studies explicitly requiring calibration exhibited the highest mean absolute Z-scores (SBP: 1.89; DBP: 1.19) together with wide CIs, suggesting that calibration does not consistently reduce error. Calibration-free methods show substantially lower values (SBP: 1.04; DBP: 0.91) with comparatively narrower intervals, indicating more robust and stable accuracy in practice. A large fraction of studies did not specify calibration requirements (NS) and reported similarly low mean absolute Z-scores (SBP: 0.78; DBP: 0.80), comparable to explicitly calibration-free methods. When NS cases were conservatively reclassified as calibration-free, the combined group maintained low error levels (SBP: 0.86; DBP: 0.84).

These results demonstrate that calibration-free models are not only feasible but, in many cases, outperform calibrated approaches, which often suffer from larger variability despite their more complex design. The fact that unspecified studies align closely with the calibration-free group further suggests that calibration requirements are gradually diminishing in contemporary cuffless and database-based research, a trend consistent with the drive toward user-friendly and scalable BP monitoring solutions.

I. Evaluation of Estimation Approaches

To evaluate the influence of estimation techniques on performance, we compared studies using deterministic, ML, and DL approaches (Table VIII).

Among the reported metrics, mean absolute Z-scores were used as the primary measure for comparing method categories. Deterministic approaches yielded the lowest mean absolute Z-scores (SBP 0.86; DBP 0.73) with the narrowest CIs, indicating stable performance. DL methods showed moderate values (SBP 1.23; DBP 1.12). ML methods exhibited the highest variability, with the largest mean absolute Z-scores

TABLE VIII
MEAN Z-SCORES AND 95% CI FOR ESTIMATION APPROACHES

Estimation Approaches	Z-score (Mean)	Z-score (Mean)	95% CI	N
SBP	Deterministic	0.81	0.86 (0.48 to 1.15)	21
	ML	1.62	1.62 (0.35 to 2.88)	19
	DL	1.20	1.23 (0.57 to 1.83)	36
DBP	Deterministic	0.70	0.73 (0.42 to 0.97)	20
	ML	1.40	1.41 (0.31 to 2.50)	17
	DL	1.12	1.12 (0.47 to 1.78)	34

(SBP 1.62; DBP 1.41) and the widest CIs, reflecting substantial heterogeneity among studies.

VI. DISCUSSION

Numerous review articles have already been published on methods for measuring BP, each with a different methodological focus and scope. To provide a clearer overview of these differences, Table IX summarizes how our work compares with previous surveys. For instance, the review by Pratiwi et al. [13] concentrates on automated auscultatory methods, while Lu et al. [14] narrow their attention to video-based BP estimation. Barvik et al. [15] and Le et al. [16] primarily emphasize continuous monitoring strategies, whereas Argha et al. [17] and Sheoran et al. [18] focus specifically on AI approaches. These reviews provide valuable in-depth analyses of their chosen domains, but their coverage is necessarily limited.

By contrast, our review adopts a broader integrative perspective, spanning cuff-based, cuffless, and contactless methods within a single framework and systematically incorporating developments from 2020 to early 2025. This approach differentiates it from both narrower reviews and broader surveys such as those by Panula et al. [167] and Jegan et al. [168], which also cover multiple methods but without the same level of systematic stratification or quantitative synthesis. Similarly, the recent work by Manga et al. [169] takes a wide physiological view by including both invasive and noninvasive pressure monitoring (blood, intracranial, pulmonary, and hepatic) with AI integration, but it does not provide a method-specific quantitative comparison for BP alone.

Specifically, the novel contributions of this review include the following.

- 1) Inclusion of standardized accuracy comparison tables and Z-score analyses, enabling direct, quantitative cross-study evaluation across heterogeneous datasets and error metrics.
- 2) Integration of both qualitative and quantitative perspectives, combining strengths, weaknesses, opportunities, and threats (SWOT) analysis, ethical and practical considerations with stratified numerical evaluation by study quality, calibration requirement, and methodological approach.
- 3) A unified dual-perspective framework that bridges statistical rigor with conceptual synthesis, offering insights into both clinical applicability and technological maturity.
- 4) The review offers a broader scope and methodological transparency than previous reviews, encompassing cuff-based, cuffless, and contactless methods with clear differentiation and structured comparison.

We acknowledge that, compared to some highly specialized surveys, our review devotes less detail to certain aspects

TABLE IX
COMPARISON WITH PREVIOUS REVIEW ARTICLES ON NONINVASIVE BP MEASUREMENT

Review (Year)	Focus / Scope	Methodological Characteristics	Time Span Covered	Unique Contribution	Limitations
Pratiwi et al. [13] (2021)	Automated auscultatory techniques	Focused on Korotkoff sounds, auscultation-based estimation.	2002-2020	Detailed analysis of automated auscultatory approaches.	Narrow scope; does not cover cuffless/contactless methods.
Lu et al. [14] (2020)	Video-based (rPPG) blood pressure estimation	Camera/video methods, non-contact.	2015-2019	Overview of video-based BP determination.	Limited to one modality, with no broader methodological comparison.
Barvík et al. [15] (2022)	Continuous monitoring, calibration models	Discussion of calibration needs, sensor technologies, algorithms for continuous monitoring, and methodological review.	NS	Detailed review of calibration models for continuous BP estimation.	Narrow scope; does not address non-continuous methods.
Le et al. [16] (2020)	Continuous non-invasive BP monitoring	Summarizes sensor modalities (PPG, ECG), physiological bases, algorithms, and methodological review.	NS	Overview of continuous monitoring approaches.	Limited scope, no comparison with cuffed/contactless.
Argha et al. [17] (2022)	AI-driven BP estimation (auscultatory + oscillometric)	Methodological review of ML and DL applied to cuff-based signals; analysis of validation standards.	Covers works up to ~2020	Methodological review of AI applied to oscillometric/auscultatory data.	Does not include cuffless/contactless or multimodal approaches.
Sheoran et al. [18] (2022)	AI-based non-invasive BP monitoring	Review of AI (ML/DL) approaches to various cuffless modalities.	NS	Summarizes AI-based approaches.	Less methodological depth; limited quantitative comparison.
Panula et al. [162] (2021)	Advances in non-invasive BP measurement (wide range of methods)	Covers contact-based and contactless technologies and wearable sensors.	NS	Relatively wide scope, including novel sensors and modalities.	Not as comprehensive in systematic comparison as this review.
Jegan et al. [163] (2022)	Review of non-invasive BP estimation via modern approaches	Focus on biomedical signals (PPG, ECG), cuffless monitoring, and ML integration.	NS	Summarizes biosignal-based methods (PTT, PAT, PWV) and the role of ML.	Mainly descriptive, it lacks systematic evaluation or comparison.
Manga et al. [164] (2023)	Advances in invasive and non-invasive BP measurement; AI integration	Review of invasive and non-invasive methods for multiple pressures (intracranial, pulmonary, hepatic portal) and AI applications across modalities.	NS	Broad physiological scope; integration of AI approaches; highlights novel microwave sensing	Narrative synthesis; no systematic criteria; lacks quantitative comparison and focus on calibration/validation
This review (2025)	Cuffed, cuffless, and contactless methods; AI integration	Systematic approach with clear inclusion/exclusion criteria; combines qualitative synthesis with quantitative comparison.	2020–2024	Broadest scope; integrates accuracy tables, standardized error comparison, and quality assessment across studies.	Less focus on calibration and study quality. Limited analysis of usability and wearables vs. non-wearables.

such as calibration strategies [15] or the distinction between wearable and nonwearable devices. However, we see this as a deliberate tradeoff. By prioritizing comprehensive coverage across all noninvasive approaches and coupling it with quantitative and qualitative synthesis, our review contributes a unique, integrative reference point for future research in the field.

A. Discussion of Study Outcomes

Our analysis confirms that traditional cuff-based methods (Table I) remain the most reliable reference for BP measurement, primarily due to their established accuracy and regulatory acceptance. Interestingly, recent work such as that of Argha et al. [21] and Argha and Celler [30], which integrated DL into oscillometric and auscultatory signals, demonstrates that even these mature techniques can benefit from modern signal processing. This finding highlights a larger trend: cuff-based devices are not likely to be replaced in clinical practice anytime soon, but using advanced computational models may improve their performance and keep them useful in a time when cuffless and contactless technologies are becoming more common. Despite their reliability, cuff-based devices remain limited by their intermittent nature and discomfort during inflation, which restricts their suitability for continuous or wearable monitoring.

Cuffless methods utilizing direct contact sensors (Table II) demonstrate clear promise, particularly when multiple signals such as PPG and ECG are combined, which consistently improves performance over single-signal approaches. Their main advantage lies in patient comfort and the ability to enable continuous monitoring at different body sites, positioning them as a highly attractive alternative for wearable applications. Nevertheless, the heterogeneity of reported metrics and study conditions complicates direct comparison across studies, underscoring the lack of standardization in this field. A persistent limitation is the dependence on calibration, which, despite being thoroughly examined in Section VI-E, continues to be a significant obstacle to clinical implementation. It is also noteworthy that while AI-driven models generally outperform deterministic approaches, examples such as Marzorati et al. [79] illustrate that carefully designed traditional methods can still yield competitive accuracy, reminding us that algorithmic sophistication alone does not guarantee superiority. Importantly, almost all studies in this category were conducted under static conditions, with patients at rest. Only two investigations [39], [86] explicitly addressed dynamic scenarios involving movement. Remarkably, their results remained comparable to static studies, suggesting that, with appropriate signal processing, cuffless methods can mitigate motion artifacts more effectively than often assumed.

Database-derived methods (Table III) provide useful benchmarking opportunities, and studies integrating both PPG and ECG signals often achieve the most consistent accuracy. While single-signal PPG approaches can in some cases reach comparable results, their higher variability underscores the fragility of relying on a single modality. This contrast highlights an important tradeoff between methodological simplicity and robustness, which is particularly relevant when designing systems for real-world deployment. Interestingly, performance appeared to depend more on the signal processing methodology than on the specific database employed, suggesting that algorithmic design plays a greater role than the choice of dataset. However, as most database-derived studies remain detached from clinical control, their results should be viewed primarily as proof of concept rather than as evidence of clinical readiness.

Despite being at the forefront of innovation in BP monitoring, contactless methods (Table IV) still lack the accuracy of contact-based approaches. While selected implementations, such as Shaikh et al.'s [142] IR camera system, have demonstrated impressive performance, most studies reported substantially higher MEs compared to cuffless or cuff-based techniques. Among the different contactless modalities, rPPG approaches generally outperformed radar in terms of stability, as reflected by lower variability, though still not reaching the reliability required for clinical adoption. This contrast illustrates both the promise and fragility of contactless monitoring; it offers unmatched comfort and unobtrusiveness, but its technical maturity is not yet sufficient for deployment beyond research settings. Moreover, the very unobtrusiveness that makes these methods appealing also raises ethical questions about consent and privacy, which are addressed in detail in Section VI-H.

B. Discussion of Comparative Results

The comparative analysis of standardized errors (Table V) provides a consolidated view of performance across method categories. Cuff-based systems exhibited the lowest mean absolute Z-scores for both systolic (0.56) and diastolic (0.56) BP, with narrow CIs, reflecting their superior consistency and reliability. These findings reinforce the role of cuff-based devices as the clinical benchmark, supported by decades of validation and standardized protocols. Their performance advantage is partly attributable to the controlled and well-established nature of oscillometric and auscultatory techniques, which reduces variability across studies.

Nevertheless, cuff-based methods also present well-known drawbacks. They provide only intermittent measurements, which limit their ability to capture rapid or nocturnal BP variations. The inflation process may cause discomfort or sleep disturbance, reducing long-term compliance in ambulatory monitoring. In addition, such devices are generally unsuitable for continuous or wearable applications, and their dependence on proper cuff size and placement can introduce operator-related variability. These limitations motivate the development of cuffless and contactless alternatives, aiming for unobtrusive, continuous, and patient-friendly monitoring.

Cuffless database-derived methods followed closely, achieving relatively low mean absolute Z-scores (0.69 for SBP, 0.67

for DBP). These results suggest that AI-driven models trained on curated datasets can approximate the accuracy of cuff-based measurements. However, their external validity remains uncertain, as curated datasets often exclude real-world noise, heterogeneous populations, and uncontrolled conditions. By contrast, cuffless experimental methods showed substantially higher errors (1.55 for SBP and 1.13 for DBP), reflecting the challenges of calibration, variability in study design, and signal sensitivity in real-life measurements.

Contactless methods represented the least reliable category, with mean absolute Z-scores of 1.67 for SBP and 1.97 for DBP, along with broad CIs. This indicates pronounced error variability and reduced robustness in uncontrolled environments, where factors such as motion, lighting, or environmental interference severely degrade performance. Although some isolated studies (e.g., IR imaging or refined rPPG) achieved encouraging results, the overall evidence points to substantial technical immaturity.

These results highlight a fundamental tradeoff: cuff-based methods remain the most accurate and consistent but are limited in terms of comfort and continuous monitoring, whereas cuffless and contactless approaches offer greater usability but at the cost of increased variability and reduced reliability.

C. SWOT-Based Comparative Discussion

To complement the quantitative evaluation, the SWOT framework (Fig. 12) further illustrates the strengths and limitations of each category.

Cuff-based systems remain distinguished by their high accuracy, established clinical acceptance, and standardized validation protocols. However, their discontinuous operation limits applicability in wearable and long-term monitoring. Recent studies emphasize AI-driven calibration and uncertainty modeling as promising approaches to improve measurement repeatability and enable remote or telemedicine-based use [20], [30]. Hybrid acoustic-oscillometric designs combining Korotkoff sound analysis with pressure sensing may further mitigate observer dependency and provide internal reference channels for algorithm validation. Nevertheless, acoustic implementations remain vulnerable to environmental noise, cuff deflation dynamics, and operator variability, reducing reliability in automated contexts [19], [23]. As digital health and cuffless monitoring technologies advance, traditional cuff-based systems face increasing risks of technological obsolescence and reduced adoption among younger or mobile-health-oriented users [30].

Cuffless methods, particularly wearable implementations, provide high comfort and seamless integration with multi-modal sensing modalities such as PPG and ECG. However, calibration requirements, motion sensitivity, and interindividual variability continue to constrain their performance. These systems introduce new opportunities for unobtrusive, continuous, and telemedicine-compatible BP monitoring, especially when integrated into wearable and multisensor platforms [45], [108]. The combination of multiwavelength PPG, ECG, and PPW sensing enables more robust estimation of systolic, diastolic, and mean pressures, while DL models and personalized calibration algorithms enhance adaptability across diverse

	CUFF-BASED $ Z\text{-score}_{SBP} = 0.56$ $ Z\text{-score}_{DBP} = 0.56$	CUFFLESS $ Z\text{-score}_{SBP(e)} = 1.55$ $ Z\text{-score}_{DBP(e)} = 1.13$ $ Z\text{-score}_{SBP(d)} = 0.69$ $ Z\text{-score}_{DBP(d)} = 0.67$	CONTACTLESS $ Z\text{-score}_{SBP} = 1.67$ $ Z\text{-score}_{DBP} = 1.97$
S Strengths	Clinically validated and widely used High accuracy and reliability Standardized protocols (AAMI/ESH)	Suitable for wearable devices Continuous, comfortable monitoring Multimodal fusion advantage Near-cuff accuracy (database)	Maximum user comfort Fully unobtrusive monitoring Encouraging small-cohort IR/rPPG results
W Weaknesses	Inconvenience with repeated use Inflation discomfort persists Limited wearable suitability	Calibration often required High sensitivity to motion artefacts Variable accuracy between individuals	Highest error variability Strong environmental dependence Limited clinical validation
O Opportunities	Reference channel for validation Home telemedicine integration AI/algorithmic improvements	Developing calibration-free models Personalized telemedicine AI-assisted accuracy gains Wearable personalization	Smart-home / eldercare Population screening Multimodal sensor fusion AI-based noise robustness
T Threats	Obsolescence risk Lower acceptance in younger users	Fragmented validation protocols Slow clinical adoption Dataset overfitting risk	Privacy / consent concerns Regulatory acceptance hurdles Real-world deployment

Fig. 12. SWOT analysis of cuff-based, cuffless, and contactless noninvasive BP methods. Header values report mean absolute standardized errors ($|Z|$) aggregated across studies (Section V). For cuffless, e denotes experimental and d denotes database-derived. Strengths/weaknesses highlight accuracy, calibration, and robustness; opportunities/threats cover deployment, privacy, and regulation.

users [81], [111]. Smartphone- and smartwatch-based implementations further improve accessibility and user engagement, potentially supporting long-term hypertension management. Despite these advantages, calibration stability remains a major limitation: model accuracy often degrades over time and across heterogeneous populations, particularly among elderly or hypertensive individuals. Additionally, physiological variability, skin tone differences, motion artifacts, and inconsistent sensor contact continue to affect measurement reliability [108]. The absence of universally accepted reference channels and standardized validation protocols further complicates regulatory approval and delays clinical adoption [45], [73].

Contactless methods provide unmatched user comfort and unobtrusiveness, offering strong potential for remote and infection-safe BP monitoring. However, their limited robustness, high error variability, and lack of clinical validation currently outweigh their advantages. These systems are particularly relevant for telemedicine and pandemic scenarios, where contact-free assessment reduces infection risk and improves measurement convenience. Integration of DL techniques has demonstrated improved resilience to environmental noise and enhanced accuracy under controlled conditions [126], [132]. Smartphone- and camera-based implementations further

enable large-scale remote health monitoring and multimodal physiological assessment beyond BP alone [137]. Nonetheless, environmental sensitivity remains a key limitation; illumination, motion, and skin reflectance variations significantly affect measurement stability in uncontrolled settings. The absence of standardized datasets and validation frameworks hampers fair algorithm comparison and delays regulatory approval [123], [140]. Moreover, radar- and vision-based systems face persistent privacy concerns and multiuser interference issues, limiting their scalability and clinical adoption.

Overall, the synthesis of quantitative and qualitative perspectives underscores that cuff-based systems remain the gold standard, while cuffless methods are rapidly approaching comparable accuracy. With further standardization, robust calibration strategies, and large-scale real-world validation, cuffless technologies may evolve into a clinically viable alternative, whereas contactless monitoring still requires significant methodological and technical progress before clinical adoption can be realized.

From these insights, several actionable recommendations can be drawn.

1) Adapt and extend existing validation protocols (e.g., AAMI/ESH/ISO) to better address cuffless and contactless

technologies, with emphasis on ambulatory and diverse population scenarios.

2) Promote multimodal sensor fusion and adaptive algorithms to mitigate noise, motion artifacts, and interindividual variability.

3) Support the development of cost-effective and power-efficient hardware to enable long-term wearable and home-based monitoring.

4) Ensure strict adherence to existing ethical and legal frameworks [e.g., general data protection regulation (GDPR)] while translating them into practical guidelines for privacy, informed consent, and user acceptance in unobtrusive monitoring systems.

D. Discussion of Study Quality

The quality assessment of the included studies (Table VI) revealed a heterogeneous but informative distribution. None of the reviewed works fell into the category of very low quality, and only a few were classified as low quality. The majority of studies clustered around medium quality, followed by a smaller proportion of high quality works, while fewer than ten studies achieved the very highest quality scores. Within this latter group, those reaching the maximum score of 10 fully complied with international validation standards, including adequate sample size, appropriate gender balance, and error margins within protocol thresholds. Studies with a score of 9 typically fell short by only a small margin, such as recruiting slightly fewer than the required number of participants, imbalanced sex ratios, or marginally exceeding acceptable error thresholds.

Interestingly, the relationship between study quality and reported error is not linear. Some lower quality studies exhibited deceptively low errors, which may partly reflect their small sample sizes and limited transparency regarding measurement conditions. By contrast, medium and high quality studies generally provided more reliable results, even though their reported errors were sometimes higher. This trend is particularly evident among medium-quality studies, many of which relied on cuffless database-derived data. These studies often demonstrated strong performance, but their overall quality score was reduced because they lacked control over data acquisition, did not involve real-time measurements, and were unable to validate their findings under clinical supervision.

High quality studies showed similarly robust performance, though they often fell short of the very highest standards due to constraints such as limited sample size or narrow participant demographics. A recurring limitation was the predominance of healthy volunteers tested under laboratory conditions rather than in diverse clinical populations. Such settings, while useful for proof of concept, reduce external validity and limit the generalizability of results to patients with comorbidities or to real-world environments with uncontrolled variability.

Paradoxically, the very highest quality studies tended to report the largest errors. This outcome can be explained by their methodological rigor, which included larger and more heterogeneous populations, clinically validated settings, and stricter adherence to validation protocols. These conditions inevitably expose greater variability than small, homogeneous

laboratory samples, but they also yield results that more accurately reflect real-world clinical performance. Thus, while their apparent accuracy may appear lower, these studies arguably provide the most trustworthy evidence base for clinical adoption.

E. Discussion of Calibration Requirements

Calibration in the context of cuffless BP monitoring refers to the process of aligning the device's measurements with a reference standard, typically by recording one or more conventional cuff-based readings and using them to adjust or train the model. This step is often assumed to improve accuracy, yet our analysis reveals a more nuanced picture.

As shown in Table VII, studies explicitly using calibration surprisingly reported higher standardized errors than those that operated without it. Even more striking, the best results were achieved by studies that did not specify whether calibration was applied. On closer inspection, most of these "unspecified" studies belonged to the cuffless, database-derived category, where calibration was effectively replaced by data-driven approaches such as DL. By training models on large datasets, these methods appear to compensate for the absence of explicit calibration and achieve lower errors through representation learning.

These findings highlight both the promise and the current ambiguity of calibration in cuffless monitoring. On the one hand, calibration can correct for individual physiological differences and device drift, but it also risks overfitting to specific conditions or becoming impractical if frequent recalibration is required. On the other hand, DL models trained on large and diverse datasets may inherently capture interindividual variability, reducing or even eliminating the need for explicit calibration. However, this advantage comes at the cost of transparency: it is not always clear whether good performance reflects genuine robustness or simply dataset-specific optimization.

In practice, the absence of consistent reporting on calibration across studies further complicates the interpretation of results. Without standardized calibration protocols, it is difficult to compare findings or assess their clinical relevance. Future work should therefore aim to establish clear guidelines on when and how calibration should be performed and whether data-driven methods can reliably replace traditional recalibration procedures in real-world applications.

F. Discussion of Estimation Approaches

The comparative analysis of methodological approaches (Table VIII) yielded several noteworthy insights. Deterministic techniques, such as regression models or analytical transformations, achieved the best overall performance across studies. Their main strength lies in transparency: they rely on explicit physiological relationships, making their predictions understandable and relatively easy to validate clinically. This clarity enhances user trust and regulatory acceptance. However, deterministic approaches are limited by their flexibility; they often struggle to capture complex nonlinear relationships and may fail under conditions that deviate from their underlying

TABLE X
HORIZONTAL COMPARISON OF ML AND DL MODEL CATEGORIES IN BP ESTIMATION

Category / Criterion	Data amount to train	Data complexity	Robustness to noise / artefacts	Interpretability	Computational complexity	Adaptability / multimodal fusion	Calibration independence
ML	Tree-based / Ensemble	++	++	++	++	++	++
	Kernel-based	++	++	++	++	++	++
	Bayesian / Probabilistic	+	+	+++	+++	+	++
	Linear / Regression	+++	+	+++	+++	+	++
	Nonlinear / Advanced	++	++	++	++	++	++
DL	Feedforward	++	++	++	++	++	++
	Convolutional	+	+++	+	++	++	+++
	Recurrent / Temporal	+	+++	+	+	+++	+++
	Transformer	+	+++	+	+	+++	+++
	Generative / Adaptation	+	+++	+	+	+++	+++

+ indicates low capability or weakest performance.

+++ indicates high capability or strongest performance for the given criterion.

The symbols represent relative capability (i.e., suitability or effectiveness) rather than computational cost or resource demand.

assumptions, such as arrhythmias, arterial stiffness variability, or motion artifacts.

ML methods performed the worst overall, reflecting their dependence on limited feature sets, small datasets, and manual feature engineering. ML algorithms such as decision trees or SVMs can adapt better than purely deterministic models, but their performance strongly depends on the choice of input features and parameter tuning. Inconsistent feature definitions and heterogeneous data sources across studies further compound these weaknesses, which explains the observed variability and reduced accuracy.

DL methods occupied an intermediate position, generally outperforming classical ML but not consistently surpassing deterministic models. Their strength lies in automatic feature extraction and the ability to model highly nonlinear dependencies, which makes them particularly attractive for multimodal data integration (e.g., combining PPG, ECG, and radar). DL models have shown promise in mitigating motion artifacts and adapting to heterogeneous signals, especially in controlled, experimental, or database settings. At the same time, they introduce significant challenges: a risk of overfitting when trained on small or homogeneous datasets, heavy reliance on large-scale annotated data, high computational costs, and limited interpretability. These drawbacks complicate their clinical translation, as black-box models struggle to provide physiological explanations acceptable to medical practitioners.

From a broader perspective, data-driven methods offer unique opportunities to advance BP monitoring by exploiting multimodal signals, adapting to individual variability, and potentially eliminating the need for explicit calibration. Their weaknesses, however, cannot be overlooked: overfitting, dependence on training distribution, and lack of transparency remain significant barriers. A horizontal comparison across AI model categories (Table X) illustrates that no single approach dominates in all scenarios. For example, CNNs excel at spatial feature extraction (e.g., from imaging or waveform morphology), recurrent networks (RNNs, LSTMs, and GRUs) are more suitable for temporal dynamics (e.g., continuous monitoring under ambulatory conditions), while transformer-based architectures show promise for long-range dependencies and multimodal fusion. By contrast, deterministic methods, though less flexible, remain valuable in settings requiring

simplicity, transparency, and low computational demand, such as portable devices or environments with strict regulatory oversight.

In summary, deterministic models currently provide the most consistent accuracy, but data-driven approaches, particularly DL, are indispensable for addressing future challenges such as motion robustness, multimodal fusion, and personalized medicine. Their successful translation into clinical practice will depend on achieving a balance between accuracy, interpretability, and standardization, ensuring that algorithmic sophistication translates into genuine clinical benefit.

G. Reproducibility and Replicability Across Modalities

While this review evaluates the reliability and generalizability of current findings, it is equally important to examine how easily these studies can be replicated or reproduced across different noninvasive methods for measuring BP, especially as these technologies progress from controlled laboratory settings toward clinical and consumer applications.

Cuff-based methods generally demonstrate the highest reproducibility due to standardized measurement protocols, commercially available equipment, and well-defined calibration and validation procedures. Studies employing oscillometric or auscultatory techniques often include detailed information about device configuration, environmental conditions, and subject preparation, allowing experiments to be reliably reproduced under comparable settings.

Cuffless experimental methods, especially those based on wearable PPG and ECG sensors, show moderate replicability. Many studies provide device specifications and signal processing workflows. However, incomplete calibration protocols, lack of detail on sensor placement, and insufficient reporting of hardware/software versions often limit reproducibility. Furthermore, factors such as intrasubject variability, environmental noise, and signal preprocessing decisions (such as filtering or segmentation) can significantly impact the outcomes, making them less controllable.

Cuffless database-driven studies demonstrate strong reproducibility at the algorithmic level, thanks to the growing availability of open-access datasets. These resources make it possible to retrain and benchmark models under comparable conditions, fostering progress toward standardized evaluation.

However, the extent to which results can be fully replicated often depends on how these datasets are utilized. In many cases, studies employ only selected subsets of records or signal segments, and the specific inclusion criteria or data partitioning strategies are not always clearly described.

Contactless methods currently exhibit the lowest replicability among all modalities. Most studies rely on custom hardware and nonstandardized environments, where factors such as illumination, camera distance, and subject motion vary substantially. For instance, rPPG accuracy depends on factors such as camera resolution, frame rate, and lighting spectra, whereas radar performance is influenced by antenna orientation and electromagnetic interference. Because these parameters are rarely reported or standardized, reproducing results without identical setups remains challenging, despite the high potential of contactless sensing technologies.

To improve reproducibility across modalities, future studies should prioritize transparent documentation and consistent data handling practices. We particularly recommend the following.

- 1) Standardized device documentation, including sensor specifications, placement diagrams, sampling rates, and environmental conditions.

- 2) Open-source code and signal processing workflows, allowing independent verification and model retraining.

- 3) Clear reporting of dataset usage, specifying which records, signal segments, or data partitions were employed in analysis to enable accurate replication and fair comparison across studies.

A cross-modality comparison of replicability thus reveals that, despite notable progress, only cuff-based and some cuffless methods currently support robust and reproducible validation. Contactless approaches, while promising, remain highly dependent on specific hardware and environmental conditions, suggesting that they require greater methodological transparency and standardized reporting.

H. Ethical and Legal Considerations

The rapid emergence of cuffless and contactless BP monitoring raises not only technical but also profound ethical and legal challenges. Unlike conventional cuff-based devices, which are already standardized and clinically validated, novel sensor technologies, particularly radar and optical systems, collect highly sensitive physiological data that extend far beyond BP itself.

- 1) *Validation and Safety:* According to ISO 81060-2:2018 [170] and ISO 81060-3:2022 [171], contact devices must undergo rigorous clinical validation for intermittent and continuous use, respectively. For contactless technologies, IEEE 1708-2014 [172] and its amendment IEEE 1708a-2019 [173] require testing under realistic conditions and across heterogeneous populations. Complementary standards such as ISO 14971 (risk management for medical devices) [174] and ISO/IEC 27001 (information security) [175] underline the need to integrate safety and cybersecurity considerations from the outset. Ensuring compliance with these standards is not merely regulatory but also an ethical imperative, as inaccurate or unsafe devices directly undermine the principle of nonmaleficence.

- 2) *Privacy and Data Protection:* With the advent of contactless sensing, privacy becomes a central concern. Radar- or camera-based systems can capture biometric patterns, including respiration or heart rate, sometimes without a subject's awareness. Under the EU's GDPR (Regulation (EU) 2016/679) [176], such data belong to a special category requiring explicit consent, strict minimization, encryption, and often a Data Protection Impact Assessment (DPIA). Risks extend beyond obvious data breaches, including metadata analysis or secondary use of information for profiling. Ethically responsible deployment, therefore, requires not only compliance with GDPR but also proactive safeguards such as privacy-preserving AI (federated learning and differential privacy) and audit trails documenting data access.

- 3) *AI Transparency and Autonomy:* Camera-based and AI-driven methods illustrate the tension between usability and accountability. Black-box DL models may yield accurate estimates but undermine physician and patient trust if the reasoning remains opaque. Integrating explainable AI, independent audits of training data, and mechanisms for human oversight are essential to ensure that technological advancement does not erode patient autonomy. At the same time, continuous monitoring must not lead to the "medicalization of everyday life," where patients feel pressured into constant self-surveillance. Protecting autonomy requires that patients retain clear rights to informed consent, refusal of monitoring, and access to their personal data in machine-readable formats.

- 4) *Clinical Integration:* Finally, the transition of innovative technologies into healthcare must follow Good Clinical Practice (ISO 14155:2020 [177]), clearly defining the role of device-generated data in medical decision-making. Without this, there is a danger that unverified signals from consumer wearables or home-use devices may influence clinical decisions without adequate professional validation.

Altogether, an ethically responsible framework for BP monitoring combines clinical accuracy, ISO/IEEE-based validation, GDPR-compliant data handling, transparent and auditable AI, and respect for patient autonomy. Only by aligning technical innovation with these ethical and legal safeguards can contactless and cuffless systems gain the trust required for sustainable integration into healthcare.

I. Practical Usability and Real-World Deployment

Although accuracy and methodological rigor dominate most studies on noninvasive BP monitoring, far fewer investigations explicitly address practical considerations such as cost, durability, power consumption, or patient acceptance. This represents a critical knowledge gap, as clinical adoption ultimately depends not only on accuracy but also on usability and sustainability in real-world settings.

Cuff-based devices remain the most accessible and cost-effective option, with established acceptance for both home and clinical use. Their main limitations relate to comfort and the discontinuity of measurements, which restrict long-term or ambulatory monitoring. Nevertheless, their relatively low cost, long battery life, and straightforward operation ensure their continuing dominance in routine practice.

Cuffless wearable devices, particularly PPG-based sensors, offer improved comfort and the potential for continuous

monitoring. Patients are increasingly accepting these devices, especially when they integrate them into their daily lives. However, their higher costs, need for calibration, and sensitivity to motion artifacts currently limit scalability. Importantly, few studies have reported on critical factors such as device durability, battery autonomy, or the robustness of wireless data transmission—issues that are central to large-scale deployment in home or outpatient settings.

Contactless methods, such as rPPG and radar, promise maximal convenience and scalability, with smartphone-based solutions being particularly attractive in low-resource environments. Nonetheless, their dependency on environmental conditions (lighting, positioning, or radio interference) and unresolved privacy concerns makes them less practical for reliable clinical deployment at present. Radar-based solutions in particular face barriers of high cost, high power consumption, and limited portability, confining their use to research or specialized clinics rather than everyday care.

Overall, the absence of systematic reporting on practical metrics such as cost, energy requirements, device lifespan, and patient adherence limits our ability to compare methods beyond accuracy alone. This highlights an urgent need for standardized frameworks to evaluate usability, scalability, and cost-effectiveness alongside clinical accuracy. Without this broader perspective, even technically promising solutions may fail to achieve meaningful impact on real-world healthcare.

J. Cross-Domain Technological Parallels

Recent advancements in cuffless and contactless BP monitoring, particularly those based on PPG, ECG, radar, and rPPG, closely parallel developments in related biomedical and computational domains. These sensing modalities have simultaneously matured in applications such as sleep and respiratory monitoring, arrhythmia detection, emotion recognition, and stress assessment, sharing common challenges in signal quality, motion artifacts, and environmental variability.

For instance, PPG and ECG signals used in commercial wearables for heart rate, arrhythmia, and oxygen saturation monitoring employ acquisition and preprocessing schemes similar to those required for cuffless BP estimation. Likewise, rPPG and radar sensing, widely adopted in telemedicine, driver monitoring, and home healthcare, offer algorithmic advances in illumination control, motion compensation, and signal fusion that can directly inform BP estimation frameworks. Recognizing and leveraging these cross-domain parallels provides an opportunity to utilize validated processing pipelines, multimodal hardware platforms, and transfer learning strategies, ultimately accelerating the robustness and clinical maturity of cuffless and contactless BP technologies.

K. Limitation of This Review

Despite its broad scope, this review has several limitations that should be acknowledged. First, the inclusion of studies relied on reported methodological details, which were sometimes incomplete or inconsistent. Aspects such as calibration procedures, participant demographics, and measurement conditions were not always transparently described, limiting the

comparability of results across studies. Second, the heterogeneity of study designs and outcome metrics necessitated standardization through Z-scores, which, while enabling comparisons, may oversimplify underlying differences in study methodology and statistical reporting.

Third, most of the studies included were performed under laboratory or otherwise controlled conditions, with only a few explicitly addressing dynamic scenarios or diverse patient populations. This restricts the generalizability of findings to real-world environments and clinical practice. Fourth, practical aspects such as device cost, battery life, long-term durability, and patient adherence were rarely reported, meaning that our discussion of usability and scalability is necessarily more qualitative than quantitative.

Finally, while this review aimed to provide a comprehensive synthesis, it inevitably reflects the current state of the literature, which is itself unevenly distributed across different technologies. Cuffless and contactless methods are highly represented, whereas wearable implementations were only marginally covered in the available studies. This relative under-representation limits our ability to fully assess the future potential of wearables, even though they are likely to become central to real-world BP monitoring. Therefore, we should view the conclusions drawn here as indicative of emerging trends rather than definitive evidence of clinical readiness.

L. Future Perspectives

Looking ahead, several directions appear central to the future of noninvasive BP monitoring. Foremost is the continued development of standardization frameworks. While cuff-based devices already follow established validation protocols such as the AAMI/ESH/ISO standards [161], comparable consensus is still lacking for cuffless and contactless methods. Current studies in these domains are often exploratory in nature and therefore do not strictly adhere to standardized validation. Nonetheless, as these approaches progress toward clinical translation, harmonized protocols tailored to their specific characteristics will become essential for comparability, regulatory acceptance, and clinical trust.

To guide this transition, a tiered validation pathway can be envisioned, extending traditional cuff-based evaluation frameworks to emerging cuffless and contactless modalities. The proposed approach involves four complementary levels.

- 1) *Analytical Validation*: Verification of signal integrity, synchronization between modalities (e.g., ECG–PPG–radar), and noise robustness under controlled testing conditions.
- 2) *Controlled Laboratory Validation*: Testing on standardized populations with clinically validated reference devices to establish baseline accuracy under semi-controlled conditions.
- 3) *Ambulatory and Home Validation*: Evaluation of stability and performance under real-world conditions, including motion, posture changes, and illumination variability.
- 4) *Generalizability Assessment*: Cross-site and cross-dataset validation across demographic subgroups (age, sex, skin tone, and comorbidities) to confirm fairness and robustness.

Collectively, this multistage framework bridges current research practices with regulatory expectations and clinical usability, providing a clear pathway from prototype evaluation to clinical implementation.

Equally important is the need for greater diversity of populations and environments. The majority of current studies rely on healthy volunteers under laboratory conditions, whereas real-world monitoring must contend with heterogeneous populations, comorbidities, and uncontrolled environments. Expanding research into these domains is essential to ensure that novel technologies deliver robust performance where they are most needed. This also includes addressing fairness in AI-driven methods, ensuring that predictive models perform consistently across age groups, sexes, and ethnicities to avoid systematic bias.

Technologically, multimodal sensing and wearable integration are likely to define the next generation of monitoring solutions. Combining complementary signals (e.g., PPG, ECG, accelerometry, or even imaging data) offers a pathway to higher robustness and personalized assessment, while wearable devices provide the comfort and continuity necessary for long-term adoption. Adaptive strategies that dynamically weight modalities based on signal quality may improve resilience in real-world situations even more than simple fusion. Despite being mentioned only briefly in the current literature, wearables are likely to play a central role in the future ecosystem of BP monitoring.

Transparency and reproducibility will remain critical enablers of progress. Open-source code, shared signal processing pipelines, and clear reporting of dataset usage, including which records or signal segments were analyzed, are essential for accurate replication and fair benchmarking. Collaborative efforts to develop harmonized evaluation frameworks and publicly available multimodal datasets would further improve cross-study comparability and strengthen the overall evidence base in this field.

For cuffless and contactless systems to achieve true clinical readiness, validation must extend beyond controlled laboratory settings to demonstrate reliability, long-term stability, and usability in real-world environments. Future studies should ensure consistent performance across diverse populations while integrating ethical and regulatory considerations, particularly privacy, safety, and accountability in AI-enabled monitoring, early in system development. Ultimately, progress in this field will depend not only on technological innovation but also on the ability to show tangible clinical utility: improving hypertension detection, treatment adherence, and decision-making in everyday care. Only by aligning innovation with rigorous validation, sustained reliability, and ethical safeguards can noninvasive BP monitoring fully transition into a clinically reliable and widely adopted component of personalized healthcare.

VII. CONCLUSION

The evolution of BP monitoring technologies is transforming cardiovascular care, with noninvasive methods opening new pathways toward continuous, comfortable, and patient-centered assessment. While cuff-based devices remain the clinical gold standard due to their accuracy and established

regulatory validation, their intermittent nature and lack of wearability limit their role in long-term monitoring.

This review provides a comprehensive and up-to-date synthesis of noninvasive approaches, encompassing cuff-based, cuffless, and contactless methods. By combining a quantitative comparison using standardized Z-scores with qualitative evaluations such as SWOT analysis, study quality assessment, calibration analysis, and methodological comparisons, it offers a dual-perspective framework that integrates statistical rigor with strategic interpretation.

The findings highlight important tradeoffs. Cuff-based methods continue to provide the most consistent and validated accuracy; cuffless methods—particularly multimodal and AI-driven—show promising potential for wearable integration, while contactless solutions offer maximal comfort but remain constrained by technical and environmental variability. Deterministic models currently achieve the most robust accuracy, yet data-driven methods, especially DL, hold the greatest promise for calibration-free, multimodal, and personalized monitoring.

Beyond accuracy, future progress will depend on addressing broader challenges. These include the establishment of standardized validation protocols tailored to cuffless and contactless devices, expansion of clinical validation into diverse populations and real-world settings, assurance of long-term stability of devices and algorithms, and safeguards for privacy, transparency, and fairness in AI. Wearables, only marginally covered in current literature, are expected to play a pivotal role in the transition from laboratory concepts to scalable healthcare solutions.

In summary, noninvasive BP monitoring is undergoing a significant transition: established technologies offer dependable benchmarks, while emerging methods are swiftly progressing toward clinical applicability. With sustained innovation, interdisciplinary collaboration, and the development of standardized and ethically grounded protocols, these technologies have the potential to fundamentally alter hypertension management, expanding accessibility, personalization, and preventive cardiovascular care worldwide.

REFERENCES

- [1] S. Kim et al., "Diagnostic accuracy of manual office blood pressure measurement in ambulatory hypertensive patients in Korea," *Korean J. Internal Med.*, vol. 33, no. 1, pp. 113–120, Jan. 2018, doi: [10.3904/kjim.2016.161](https://doi.org/10.3904/kjim.2016.161).
- [2] Z.-D. Liu, J.-K. Liu, B. Wen, Q.-Y. He, Y. Li, and F. Miao, "Cuffless blood pressure estimation using pressure pulse wave signals," *Sensors*, vol. 18, no. 12, p. 4227, Dec. 2018, doi: [10.3390/s18124227](https://doi.org/10.3390/s18124227).
- [3] G. Lee et al., "Machine-learning prediction of kidney failure occurrence based on regular health check-up data: A nationwide cohort dataset in South Korea," *Life Cycle*, vol. 4, p. e6, Aug. 2024, doi: [10.54724/lc.2024.e6](https://doi.org/10.54724/lc.2024.e6).
- [4] H. M. G. Glyde, C. Morgan, T. M. A. Wilkinson, I. T. Nabney, and J. W. Dodd, "Remote patient monitoring and machine learning in acute exacerbations of chronic obstructive pulmonary disease: Dual systematic literature review and narrative synthesis," *J. Med. Internet Res.*, vol. 26, Sep. 2024, Art. no. e52143, doi: [10.2196/52143](https://doi.org/10.2196/52143).
- [5] M. Roerecke, J. Kaczorowski, and M. G. Myers, "Comparing automated office blood pressure readings with other methods of blood pressure measurement for identifying patients with possible hypertension: A systematic review and meta-analysis," *JAMA Intern. Med.*, vol. 179, no. 3, p. 351, Mar. 2019, doi: [10.1001/jamainternmed.2018.6551](https://doi.org/10.1001/jamainternmed.2018.6551).

- [6] M. Zuber, H.-H. Schäfer, W. Kaiser, and P. Erne, "Measuring accuracy of sphygmomanometers in the medical practices of Swiss primary care physicians," *Eur. J. Gen. Pract.*, vol. 19, no. 4, pp. 244–247, Dec. 2013, doi: [10.3109/13814788.2013.779664](https://doi.org/10.3109/13814788.2013.779664).
- [7] M. G. Myers et al., "Conventional versus automated measurement of blood pressure in primary care patients with systolic hypertension: Randomised parallel design controlled trial," *BMJ*, vol. 342, no. feb07 1, p. d286, Feb. 2011, doi: [10.1136/bmj.d286](https://doi.org/10.1136/bmj.d286).
- [8] B. J. Kim, S. H. Lee, M. Y. Lee, S. J. Lee, and H. I. Choi, "Comparison of office blood pressure, automated unattended office blood pressure, home blood pressure, and 24-hour ambulatory blood pressure measurements," *J. Korean Med. Sci.*, vol. 38, no. 48, p. 406, 2023, doi: [10.3346/jkms.2023.38.e406](https://doi.org/10.3346/jkms.2023.38.e406).
- [9] A. Kollias, E. Stambolliu, K. G. Kyriakoulis, A. Gravvani, and G. S. Stergiou, "Unattended versus attended automated office blood pressure: Systematic review and meta-analysis of studies using the same methodology for both methods," *J. Clin. Hypertension*, vol. 21, no. 2, pp. 148–155, Feb. 2019, doi: [10.1111/jch.13462](https://doi.org/10.1111/jch.13462).
- [10] J. K. Kim, "Blood pressure and mortality: Joint effect of blood pressure measures," *J. Clin. Cardiol. Cardiovascular Therapy*, vol. 2, no. 1, Oct. 2020, Art. no. 1009, doi: [10.31546/2633-7916.1009](https://doi.org/10.31546/2633-7916.1009).
- [11] B. Aggarwal et al., "Effects of inadequate sleep on blood pressure and endothelial inflammation in women: Findings from the American Heart Association Go Red for Women Strategically Focused Research Network," *J. Amer. Heart Assoc.*, vol. 7, no. 12, Jun. 2018, Art. no. 008590, doi: [10.1161/jaha.118.008590](https://doi.org/10.1161/jaha.118.008590).
- [12] H. Mou et al., "Using data augmentation to improve the accuracy of blood pressure measurement based on photoplethysmography," *Electronics*, vol. 13, no. 8, p. 1599, Apr. 2024, doi: [10.3390/electronics13081599](https://doi.org/10.3390/electronics13081599).
- [13] N. G. Pratiwi, A. W. Setiawan, D. Naufal, and L. Lindayani, "A review of equipment and signal processing of the automated auscultation for blood pressure measurement," in *Proc. 3rd Int. Symp. Mater. Electr. Eng. Conf. (ISMEE)*, Nov. 2021, pp. 26–31, doi: [10.1109/ISMEE54273.2021.9774036](https://doi.org/10.1109/ISMEE54273.2021.9774036).
- [14] Y. Lu, C. Wang, and M. Q.-H. Meng, "Video-based contactless blood pressure estimation: A review," in *Proc. IEEE Int. Conf. Real-Time Comput. Robot. (RCAR)*, Sep. 2020, pp. 62–67, doi: [10.1109/RCAR49640.2020.9303040](https://doi.org/10.1109/RCAR49640.2020.9303040).
- [15] D. Barvik, M. Cerny, M. Penhaker, and N. Noury, "Noninvasive continuous blood pressure estimation from pulse transit time: A review of the calibration models," *IEEE Rev. Biomed. Eng.*, vol. 15, pp. 138–151, 2022, doi: [10.1109/RBME.2021.3109643](https://doi.org/10.1109/RBME.2021.3109643).
- [16] T. Le et al., "Continuous non-invasive blood pressure monitoring: A methodological review on measurement techniques," *IEEE Access*, vol. 8, pp. 212478–212498, 2020, doi: [10.1109/ACCESS.2020.3040257](https://doi.org/10.1109/ACCESS.2020.3040257).
- [17] A. Argha, B. G. Celler, and N. H. Lovell, "Artificial intelligence based blood pressure estimation from auscultatory and oscillometric waveforms: A methodological review," *IEEE Rev. Biomed. Eng.*, vol. 15, pp. 152–168, 2022, doi: [10.1109/RBME.2020.3040715](https://doi.org/10.1109/RBME.2020.3040715).
- [18] V. Sheoran, G. Raval, S. Valiveti, and S. Parikh, "A review of non invasive blood pressure monitoring using artificial intelligence based approaches," in *Proc. Int. Conf. Artif. Intell. Comput. (ICAAIC)*, May 2022, pp. 01–06, doi: [10.1109/ICAAIC53929.2022.9793040](https://doi.org/10.1109/ICAAIC53929.2022.9793040).
- [19] N. G. Pratiwi, A. W. Setiawan, D. Naufal, and L. Lindayani, "The design of Korotkoff sound detection using amplitude parameter and oscillation beat to estimate non-invasive blood pressure," in *Proc. Int. Seminar Intell. Technol. Its Appl. (ISITIA)*, Jul. 2023, pp. 587–592, doi: [10.1109/isitia59021.2023.10221160](https://doi.org/10.1109/isitia59021.2023.10221160).
- [20] E. Alvarez Vazquez, D. Ewert, D. Jorgenson, and M. Sand, "Assessment of the uncertainty associated with two consecutive blood pressure measurements using the auscultatory method," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022, doi: [10.1109/TIM.2022.3216387](https://doi.org/10.1109/TIM.2022.3216387).
- [21] A. Argha, B. G. Celler, A. Yong, I. Rubenis, M. Butlin, and A. Avolio, "New perspectives on non-invasive blood pressure measurement," in *Proc. 45th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2023, pp. 1–4, doi: [10.1109/EMBC40787.2023.10340757](https://doi.org/10.1109/EMBC40787.2023.10340757).
- [22] A. Argha, B. G. Celler, and N. H. Lovell, "A novel automated blood pressure estimation algorithm using sequences of Korotkoff sounds," *IEEE J. Biomed. Health Informat.*, vol. 25, no. 4, pp. 1257–1264, Apr. 2021, doi: [10.1109/JBHI.2020.3012567](https://doi.org/10.1109/JBHI.2020.3012567).
- [23] D. Naufal, A. W. Setiawan, and T. L. E. Rajab, "Blood pressure measuring device based on Korotkoff sound's tapping period and frequency detection," in *Proc. Int. Seminar Intell. Technol. Its Appl. (ISITIA)*, Jul. 2020, pp. 158–163, doi: [10.1109/isitia49792.2020.9163700](https://doi.org/10.1109/isitia49792.2020.9163700).
- [24] T. Adiono, R. P. Ramadhani, C. Amadeus, and S. N. C. Sinaga, "Algorithm development for stepwise valve deflation method in blood pressure measurement," in *Proc. Asia-Pacific Signal Inf. Process. Assoc. Annu. Summit Conf. (APSIPA ASC)*, Oct. 2023, pp. 1257–1262, doi: [10.1109/apsipaasc58517.2023.10317559](https://doi.org/10.1109/apsipaasc58517.2023.10317559).
- [25] S. Karimian, M. Yazdchi, and R. Hajian, "Implementation and validation of a novel noninvasive blood pressure measurement algorithm based on oscillometric method with obtaining international hypertension standards," *J. Med. Signals Sensors*, vol. 12, no. 2, pp. 114–121, Apr. 2022, doi: [10.4103/jmss.jmss_123_21](https://doi.org/10.4103/jmss.jmss_123_21).
- [26] L. I. Bogatu, S. Turco, M. Mischi, J. Muehlsteff, and P. Woerlee, "An experimental study on the blood pressure cuff as a transducer for oscillometric blood pressure measurements," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–11, 2021, doi: [10.1109/TIM.2020.3023218](https://doi.org/10.1109/TIM.2020.3023218).
- [27] N. Yoshimura et al., "Development of a wrist sphygmomanometer with an active soft mechanism," in *Proc. IEEE Int. Conf. Mechatronics Autom. (ICMA)*, Aug. 2021, pp. 1478–1483, doi: [10.1109/ICMA52036.2021.9512680](https://doi.org/10.1109/ICMA52036.2021.9512680).
- [28] R. Hampson, R. G. Anderson, and G. Dobie, "Non-invasive radial artery blood pressure monitoring using error compensated tactile sensors and patient specific oscillometry," in *Proc. 44th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Glasgow, U.K., Jul. 2022, pp. 828–831, doi: [10.1109/EMBC48229.2022.9871598](https://doi.org/10.1109/EMBC48229.2022.9871598).
- [29] A. Najd, S. Akhoondi, Y. Khezri, and A. Khadem, "Blood pressure measurement using a Gaussian curve fitting algorithm," in *Proc. 29th Nat. 7th Int. Iranian Conf. Biomed. Eng. (ICBME)*, Iran, Dec. 2022, pp. 135–139, doi: [10.1109/ICBME57741.2022.10052845](https://doi.org/10.1109/ICBME57741.2022.10052845).
- [30] A. Argha and B. G. Celler, "Blood pressure estimation from time-domain features of oscillometric waveforms using long short-term memory recurrent neural networks," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 6, pp. 3614–3622, Jun. 2020, doi: [10.1109/TIM.2019.2941037](https://doi.org/10.1109/TIM.2019.2941037).
- [31] M. Choi and S.-J. Lee, "Oscillometry-based blood pressure estimation using convolutional neural networks," *IEEE Access*, vol. 10, pp. 56813–56822, 2022, doi: [10.1109/ACCESS.2022.3177539](https://doi.org/10.1109/ACCESS.2022.3177539).
- [32] K. Takahashi, T. Asai, and Y. Okuda, "Efficacy of a new blood pressure monitor (inflationary non-invasive blood pressure, iNIBP™): A randomised controlled study," *Anaesthesia*, vol. 75, no. 1, pp. 37–44, Jan. 2020, doi: [10.1111/anae.14850](https://doi.org/10.1111/anae.14850).
- [33] T. Ursache, G. G. Petroiu, and C. Rotariu, "Remote monitoring of blood pressure by using wireless devices," in *Proc. Int. Conf. E-Health Bioeng. (EHB)*, Nov. 2021, pp. 1–4, doi: [10.1109/EHB52898.2021.9657600](https://doi.org/10.1109/EHB52898.2021.9657600).
- [34] M. Zylinski, W. Niewiadomski, G. Cybulski, and A. Gasiorowska, "Description of the volume-clamp method of blood pressure measurements using the mathematical model of the Lamé problem," in *Proc. Comput. Cardiol. (CinC)*, vol. 498, Sep. 2022, pp. 1–4, doi: [10.22489/CinC.2022.003](https://doi.org/10.22489/CinC.2022.003).
- [35] M. Żyliński and G. Cybulski, "Verification of the assumptions of volume-clamp method for continuous blood pressure measurement in a silicone phantom," in *Proc. Comput. Cardiol. (CinC)*, vol. 498, Sep. 2022, pp. 1–4, doi: [10.22489/CinC.2022.215](https://doi.org/10.22489/CinC.2022.215).
- [36] F. Farjana, S. K. Singha, and A. A. Farabi, "Cuffless blood pressure determination using photoplethysmogram (PPG) signal based on multiple linear regression analysis," in *Proc. Int. Conf. Sci. Contemp. Technol. (ICSCT)*, Aug. 2021, pp. 1–5, doi: [10.1109/ICSCT53883.2021.9642626](https://doi.org/10.1109/ICSCT53883.2021.9642626).
- [37] T. Dave, U. Pandya, and M. Joshi, "Cuff-less blood pressure measurement from wireless ECG and PPG signals," in *Proc. IEEE Int. Symp. Smart Electron. Syst. (iSES)*, Dec. 2021, pp. 33–37, doi: [10.1109/ISES52644.2021.00020](https://doi.org/10.1109/ISES52644.2021.00020).
- [38] T. Panula, J.-P. Sirkiä, and M. Kaisti, "Miniaturization of a finger-worn blood pressure instrument," in *Proc. 43rd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Nov. 2021, pp. 7441–7444, doi: [10.1109/EMBC46164.2021.9629790](https://doi.org/10.1109/EMBC46164.2021.9629790).
- [39] P. Mansourinezhad, M. M. Ahmadi, and H. Mohammadzade, "Blood pressure estimation using a PPG signal recorded from the fingertip of a hand moving in the sagittal plane," *IEEE Sensors J.*, vol. 23, no. 15, pp. 17751–17760, Aug. 2023, doi: [10.1109/JSEN.2023.3282590](https://doi.org/10.1109/JSEN.2023.3282590).
- [40] M. Freithaler, A. Chandrasekhar, V. Dhamotharan, C. Landry, S. G. Shroff, and R. Mukkamala, "Smartphone-based blood pressure monitoring via the oscillometric finger pressing method: Analysis of oscillation width variations can improve diastolic pressure computation," *IEEE Trans. Biomed. Eng.*, vol. 70, no. 11, pp. 3052–3063, Nov. 2023, doi: [10.1109/TBME.2023.3275031](https://doi.org/10.1109/TBME.2023.3275031).

- [41] Y. Lee, J. Seo, J. Park, and H. Lee, "Analysis for calibration pre-post difference in BP estimation of galaxy watch," in *Proc. 45th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2023, pp. 1–3, doi: [10.1109/EMBC40787.2023.10340129](https://doi.org/10.1109/EMBC40787.2023.10340129).
- [42] M. Pediaditis, E. G. Spanakis, G. Zacharakis, and V. Sakkalis, "Cuffless blood pressure estimation using wrist photoplethysmography," in *Proc. 43rd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Nov. 2021, pp. 7356–7359, doi: [10.1109/EMBC46164.2021.9629544](https://doi.org/10.1109/EMBC46164.2021.9629544).
- [43] A. Poli, G. Cosoli, G. Iadarola, S. Spinsante, and L. Scalise, "Feasibility of blood pressure measurement through wearable devices: Analysis of smartwatches performance," in *Proc. IEEE Int. Symp. Med. Meas. Appl. (MeMeA)*, Jun. 2022, pp. 1–6, doi: [10.1109/MEMEA54994.2022.9856533](https://doi.org/10.1109/MEMEA54994.2022.9856533).
- [44] F. Scardulla, N. Montinaro, and L. D'Acquisto, "Blood pressure acquisitions with a prototypal PPG-based device," in *Proc. IEEE Int. Workshop Metrology Ind. 4.0 IoT*, Jun. 2022, pp. 414–418, doi: [10.1109/METROIND4.0IOT54413.2022.9831598](https://doi.org/10.1109/METROIND4.0IOT54413.2022.9831598).
- [45] Z.-D. Liu et al., "Cuffless blood pressure measurement using smartwatches: A large-scale validation study," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 9, pp. 4216–4227, Sep. 2023, doi: [10.1109/JBHI.2023.3278168](https://doi.org/10.1109/JBHI.2023.3278168).
- [46] H. Mou and J. Yu, "Transfer learning method for cuffless blood pressure estimation based on measured PPG data," in *Proc. Cross Strait Radio Sci. Wireless Technol. Conf. (CSRSWTC)*, Dec. 2022, pp. 1–3, doi: [10.1109/CSRSWTC56224.2022.10098410](https://doi.org/10.1109/CSRSWTC56224.2022.10098410).
- [47] S. Li, L. Xiong, D. Xia, and H. Zhang, "Temporal attention splitting network for non-invasive blood glucose, cuff-less blood pressure estimation," in *Proc. IEEE Smartworld, Ubiquitous Intell. Comput., Scalable Comput. Commun., Digital Twin, Privacy Comput., Metaverse, Auton. Trusted Vehicles (SmartWorld/UIC/ScalCom/DigitalTwin/PriComp/Meta)*, Dec. 2022, pp. 1272–1277, doi: [10.1109/SMARTWORLD-UIC-ATC-SCALCOM-DIGITALTWIN-PRICOMP-METAVERSE56740.2022.00188](https://doi.org/10.1109/SMARTWORLD-UIC-ATC-SCALCOM-DIGITALTWIN-PRICOMP-METAVERSE56740.2022.00188).
- [48] C. Ma et al., "SMART-BP: SEM-ResNet and auto-regressor based on a two-stage framework for noninvasive blood pressure measurement," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–18, 2024, doi: [10.1109/TIM.2023.3342856](https://doi.org/10.1109/TIM.2023.3342856).
- [49] S. Gupta, A. Singh, and A. Sharma, "PPG-based end-to-end framework for cuffless blood pressure estimation using CNN-LSTM," in *Proc. Nat. Conf. Commun. (NCC)*, Feb. 2023, pp. 1–6, doi: [10.1109/NCC56989.2023.10067922](https://doi.org/10.1109/NCC56989.2023.10067922).
- [50] D. H. Nguyen and P. C.-P. Chao, "Significant improvement in precision of real-time blood pressure prediction based on complete cycles of measured PPGs," in *Proc. IEEE Sensors*, Oct. 2023, pp. 1–4, doi: [10.1109/SENSOR556945.2023.10324952](https://doi.org/10.1109/SENSOR556945.2023.10324952).
- [51] M. A. Mehrabadi, S. A. H. Aqajari, A. H. A. Zargari, N. Dutt, and A. M. Rahmani, "Novel blood pressure waveform reconstruction from photoplethysmography using cycle generative adversarial networks," in *Proc. 44th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2022, pp. 1906–1909, doi: [10.1109/EMBC48229.2022.9871962](https://doi.org/10.1109/EMBC48229.2022.9871962).
- [52] H. Tjahjadi, K. Ramli, and H. Murfi, "Noninvasive classification of blood pressure based on photoplethysmography signals using bidirectional long short-term memory and time-frequency analysis," *IEEE Access*, vol. 8, pp. 20735–20748, 2020, doi: [10.1109/ACCESS.2020.2968967](https://doi.org/10.1109/ACCESS.2020.2968967).
- [53] F. Shoeibi, E. Najafiaghdam, and A. Ebrahimi, "Non-invasive blood pressure estimation using Poincaré plot indices of photoplethysmography signals," in *Proc. 30th Int. Conf. Electr. Eng. (ICEE)*, May 2022, pp. 951–956, doi: [10.1109/icee55646.2022.9827102](https://doi.org/10.1109/icee55646.2022.9827102).
- [54] Q. Chen, X. Chen, Y. Chen, and L. Song, "Non-invasive blood pressure detection method based on CNN-LSTM-LightGBM combination model," in *Proc. 42nd Chin. Control Conf. (CCC)*, Jul. 2023, pp. 8695–8700, doi: [10.23919/ccc58697.2023.10240350](https://doi.org/10.23919/ccc58697.2023.10240350).
- [55] P. Yao et al., "Multi-dimensional feature combination method for continuous blood pressure measurement based on wrist PPG sensor," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 8, pp. 3708–3719, Aug. 2022, doi: [10.1109/JBHI.2022.3167059](https://doi.org/10.1109/JBHI.2022.3167059).
- [56] B. A. Dong and P. M. Hoang, "Estimating continuous blood pressure from photoplethysmogram signals for non-invasive devices by convolutional neural network," in *Proc. IEEE Stat. Signal Process. Workshop (SSP)*, Jul. 2023, pp. 671–675, doi: [10.1109/SSP53291.2023.10208049](https://doi.org/10.1109/SSP53291.2023.10208049).
- [57] M. Kim, H. Lee, K.-Y. Kim, and K.-H. Kim, "Deep learning model for blood pressure estimation from PPG signal," in *Proc. IEEE Int. Conf. Metrol. Extended Reality, Artif. Intell. Neural Eng. (MetroXRINE)*, Oct. 2022, pp. 1–5, doi: [10.1109/METROXRINE54828.2022.9967606](https://doi.org/10.1109/METROXRINE54828.2022.9967606).
- [58] D. Jia and W. Yin, "Continuous blood pressure prediction based on hierarchical adaptive algorithm," in *Proc. IEEE 4th Inf. Technol., Netw., Electron. Autom. Control Conf. (ITNEC)*, vol. 1, Jun. 2020, pp. 934–938, doi: [10.1109/ITNEC48623.2020.9084725](https://doi.org/10.1109/ITNEC48623.2020.9084725).
- [59] X. Liu, M. Shao, X. Ma, and H. Kuang, "A method of predicting continuous blood pressure by PPG signal based on neural network," in *Proc. IEEE 6th Inf. Technol., Netw., Electron. Autom. Control Conf. (ITNEC)*, vol. 6, Feb. 2023, pp. 1659–1663, doi: [10.1109/ITNEC56291.2023.10082184](https://doi.org/10.1109/ITNEC56291.2023.10082184).
- [60] O. Schlesinger, N. Vigderhouse, D. Eytan, and Y. Moshe, "Blood pressure estimation from PPG signals using convolutional neural networks and Siamese network," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 1135–1139, doi: [10.1109/ICASSP40776.2020.9053446](https://doi.org/10.1109/ICASSP40776.2020.9053446).
- [61] W. Wang, P. Mohseni, K. L. Kilgore, and L. Najafizadeh, "Cuffless blood pressure estimation from photoplethysmography via visibility graph and transfer learning," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 5, pp. 2075–2085, May 2022, doi: [10.1109/JBHI.2021.3128383](https://doi.org/10.1109/JBHI.2021.3128383).
- [62] S. G. Khalid, H. Liu, T. Zia, J. Zhang, F. Chen, and D. Zheng, "Cuffless blood pressure estimation using single channel photoplethysmography: A two-step method," *IEEE Access*, vol. 8, pp. 58146–58154, 2020, doi: [10.1109/ACCESS.2020.2981903](https://doi.org/10.1109/ACCESS.2020.2981903).
- [63] A. Chakraborty, D. Goswami, J. Mukhopadhyay, and S. Chakrabarti, "Measurement of arterial blood pressure through single-site acquisition of photoplethysmograph signal," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–10, 2021, doi: [10.1109/TIM.2020.3011304](https://doi.org/10.1109/TIM.2020.3011304).
- [64] A. Chakraborty, D. Goswami, J. Mukhopadhyay, and S. Chakrabarti, "Blood pressure estimation based on photoplethysmography for personalized healthcare," *IEEE Trans. Consum. Electron.*, vol. 69, no. 4, pp. 1195–1203, Nov. 2023, doi: [10.1109/TCE.2023.3316514](https://doi.org/10.1109/TCE.2023.3316514).
- [65] S. Cheemakurthi, J. Chakka, S. M. Parida, V. N. S. Praneeth, Kinjal, and K. Debasis, "A simple user-friendly Android application for oxygen and blood pressure measurement," in *Proc. IEEE 4th Int. Conf. Cybern., Cognition Mach. Learn. Appl.*, Oct. 2022, pp. 1–6, doi: [10.1109/ICCCMLA56841.2022.9988993](https://doi.org/10.1109/ICCCMLA56841.2022.9988993).
- [66] J. Leitner, P.-H. Chiang, and S. Dey, "Personalized blood pressure estimation using photoplethysmography: A transfer learning approach," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 1, pp. 218–228, Jan. 2022, doi: [10.1109/JBHI.2021.3085526](https://doi.org/10.1109/JBHI.2021.3085526).
- [67] Mahmuda, Barkatullah, M. R. Haque, A. A. Noman, E. Haque, and F. Ahmed, "Smartphone based BP level monitoring system using DNN model," in *Proc. Int. Conf. Adv. Technol. Commun. (ATC)*, Oct. 2023, pp. 12–18, doi: [10.1109/atc58710.2023.10318933](https://doi.org/10.1109/atc58710.2023.10318933).
- [68] Y. Y. Koay, R. Bakhteri, N. I. D. Zaman, and Y. W. Hau, "Cuffless blood pressure meter with mobile application for home-care service," in *Proc. 2nd Int. Conf. Intell. Cybern. Technol. Appl. (ICICyTA)*, Dec. 2022, pp. 134–138, doi: [10.1109/ICICyTA57421.2022.10038029](https://doi.org/10.1109/ICICyTA57421.2022.10038029).
- [69] P.-Y. Ko, L.-F. Lai, T.-H. Ku, and Y.-D. Lin, "A CNN approach to predicting non-invasive blood pressure of surgical patients," in *Proc. IEEE 5th Int. Conf. Knowl. Innov. Invent. (ICKII)*, Jul. 2022, pp. 9–13, doi: [10.1109/ICKII55100.2022.9983601](https://doi.org/10.1109/ICKII55100.2022.9983601).
- [70] P. Nandi and M. Rao, "A novel CNN-LSTM model based non-invasive cuff-less blood pressure estimation system," in *Proc. 44th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2022, pp. 832–836, doi: [10.1109/EMBC48229.2022.9871777](https://doi.org/10.1109/EMBC48229.2022.9871777).
- [71] K. Louka, J. Cox, I. Tan, A. P. Avolio, M. F. O'Rourke, and M. Butlin, "An investigation of the individualized, two-point calibration method for cuffless blood pressure estimation using pulse arrival time: An historical perspective using the casio BP-100 digital watch," in *Proc. 43rd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Nov. 2021, pp. 7493–7496, doi: [10.1109/EMBC46164.2021.9630063](https://doi.org/10.1109/EMBC46164.2021.9630063).
- [72] Z. Shen, L. Liu, and X. Ding, "Bayesian model averaging for improving the accuracy of cuffless blood pressure estimation," in *Proc. 44th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2022, pp. 3981–3984, doi: [10.1109/EMBC48229.2022.9871581](https://doi.org/10.1109/EMBC48229.2022.9871581).
- [73] N. Gogiberidze et al., "Practical application of a new cuffless blood pressure measurement method," *Pathophysiology*, vol. 30, no. 4, pp. 586–598, Dec. 2023, doi: [10.3390/pathophysiology30040042](https://doi.org/10.3390/pathophysiology30040042).
- [74] S. Qiu, Y.-T. Zhang, S.-K. Lau, and N. Zhao, "Scenario adaptive cuffless blood pressure estimation by integrating cardiovascular coupling effects," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 3, pp. 1375–1385, Mar. 2023, doi: [10.1109/JBHI.2022.3227235](https://doi.org/10.1109/JBHI.2022.3227235).
- [75] A. C. Podaru and V. David, "Blood pressure estimation based on synchronous ECG and PPG recording," in *Proc. Int. Conf. Expo. Electr. Power Eng. (EPE)*, Oct. 2020, pp. 640–645, doi: [10.1109/EPE50722.2020.9305544](https://doi.org/10.1109/EPE50722.2020.9305544).

- [76] A. Chandrasekhar, M. Yavarimanes, K. Natarajan, J.-O. Hahn, and R. Mukkamala, "PPG sensor contact pressure should be taken into account for cuff-less blood pressure measurement," *IEEE Trans. Biomed. Eng.*, vol. 67, no. 11, pp. 3134–3140, Nov. 2020, doi: [10.1109/TBME.2020.2976989](https://doi.org/10.1109/TBME.2020.2976989).
- [77] K. Chen, X. Zhang, Q. Zou, and X. Wang, "Individual-based cuffless continue estimation of blood pressure measurement: Using multiple pulse transmit time," in *Proc. Int. Conf. Intell. Transp., Big Data Smart City (ICITBS)*, Jan. 2020, pp. 889–892, doi: [10.1109/ICITBS49701.2020.00196](https://doi.org/10.1109/ICITBS49701.2020.00196).
- [78] X. Chen, S. Yu, Y. Zhang, F. Chu, and B. Sun, "Machine learning method for continuous noninvasive blood pressure detection based on random forest," *IEEE Access*, vol. 9, pp. 34112–34118, 2021, doi: [10.1109/ACCESS.2021.3062033](https://doi.org/10.1109/ACCESS.2021.3062033).
- [79] D. Marzorati, D. Bovio, C. Salito, L. Mainardi, and P. Cerveri, "Chest wearable apparatus for cuffless continuous blood pressure measurements based on PPG and PCG signals," *IEEE Access*, vol. 8, pp. 55424–55437, 2020, doi: [10.1109/ACCESS.2020.2981300](https://doi.org/10.1109/ACCESS.2020.2981300).
- [80] M. Al-Hashem et al., "Blood pressure estimation using optical technique and surface fitting," in *Proc. IEEE 5th Middle East Afr. Conf. Biomed. Eng. (MECBME)*, Oct. 2020, pp. 1–5, doi: [10.1109/MECBME47393.2020.9265116](https://doi.org/10.1109/MECBME47393.2020.9265116).
- [81] K. Song, K.-Y. Chung, and J.-H. Chang, "Cuffless deep learning-based blood pressure estimation for smart wristwatches," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 7, pp. 4292–4302, Jul. 2020, doi: [10.1109/TIM.2019.2947103](https://doi.org/10.1109/TIM.2019.2947103).
- [82] O. Koçak, Z. Telatar, C. Fıçıcı, and A. Koçoğlu, "Design a cuffless blood pressure measurement system," in *Proc. Med. Technol. Congr. (TIPTEKNO)*, Nov. 2023, pp. 1–4, doi: [10.1109/tiptekno59875.2023.10359204](https://doi.org/10.1109/tiptekno59875.2023.10359204).
- [83] S. Yang, Y. Zhang, S.-Y. Cho, R. Correia, and S. P. Morgan, "Non-invasive cuff-less blood pressure estimation using a hybrid deep learning model," *Opt. Quantum Electron.*, vol. 53, no. 2, pp. 1–20, Feb. 2021, doi: [10.1007/s11082-020-02667-0](https://doi.org/10.1007/s11082-020-02667-0).
- [84] U. Senturk, K. Polat, and I. Yucedag, "A non-invasive continuous cuffless blood pressure estimation using dynamic recurrent neural networks," *Appl. Acoust.*, vol. 170, Dec. 2020, Art. no. 107534, doi: [10.1016/j.apacoust.2020.107534](https://doi.org/10.1016/j.apacoust.2020.107534).
- [85] B. De Marchi, M. Frigerio, S. De Nadai, G. Longinotti-Buitoni, and A. Aliverti, "Blood pressure continuous measurement through a wearable device: Development and validation of a cuffless method," *Sensors*, vol. 21, no. 21, p. 7334, Nov. 2021, doi: [10.3390/s21217334](https://doi.org/10.3390/s21217334).
- [86] C. Landry, E. T. Hedge, R. L. Hughson, S. D. Peterson, and A. Arami, "Accurate blood pressure estimation during activities of daily living: A wearable cuffless solution," *IEEE J. Biomed. Health Informat.*, vol. 25, no. 7, pp. 2510–2520, Jul. 2021, doi: [10.1109/JBHI.2021.3054597](https://doi.org/10.1109/JBHI.2021.3054597).
- [87] A. Paviglianiti, V. Randazzo, E. Pasero, and A. Vallan, "Noninvasive arterial blood pressure estimation using ABPNet and VITAL-ECG," in *Proc. IEEE Int. Instrum. Meas. Technol. Conf. (I2MTC)*, May 2020, pp. 1–5, doi: [10.1109/I2MTC43012.2020.9129361](https://doi.org/10.1109/I2MTC43012.2020.9129361).
- [88] S. He, H. R. Dajani, and M. Bolic, "Novel cuffless blood pressure estimation method using a Bayesian hierarchical model," in *Proc. 43rd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Nov. 2021, pp. 898–901, doi: [10.1109/EMBC46164.2021.9629594](https://doi.org/10.1109/EMBC46164.2021.9629594).
- [89] J. Ji, M. Dong, Q. Lin, and K. C. Tan, "Noninvasive cuffless blood pressure estimation with dendritic neural regression," *IEEE Trans. Cybern.*, vol. 53, no. 7, pp. 4162–4174, Jul. 2023, doi: [10.1109/TCYB.2022.3141380](https://doi.org/10.1109/TCYB.2022.3141380).
- [90] C. El Hajj and P. A. Kyriacou, "Recurrent neural network models for blood pressure monitoring using PPG morphological features," in *Proc. 43rd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Nov. 2021, pp. 1865–1868, doi: [10.1109/EMBC46164.2021.9630319](https://doi.org/10.1109/EMBC46164.2021.9630319).
- [91] D. C. Martínez-Reyes, "Algorithm for the joint analysis of the ECG and PPG signals: Algoritmo para el análisis en conjunto de las señales del ECG y PPG," *UIS Ingenierías*, vol. 20, no. 4, pp. 45–58, Oct. 2021, doi: [10.18273/revuin.v20n4-2021004](https://doi.org/10.18273/revuin.v20n4-2021004).
- [92] F. Tabei, J. M. Gresham, B. Askarian, K. Jung, and J. W. Chong, "Cuff-less blood pressure monitoring system using smartphones," *IEEE Access*, vol. 8, pp. 11534–11545, 2020, doi: [10.1109/ACCESS.2020.2965082](https://doi.org/10.1109/ACCESS.2020.2965082).
- [93] N. J. Kewalramani, K. A. Srivastava, J. P. Kalambe, and R. R. Khandelwal, "Monitoring of blood pressure with video processing," in *Proc. 14th Int. Conf. Comput. Commun. Netw. Technol. (ICCCNT)*, Jul. 2023, pp. 1–5, doi: [10.1109/icccnt56998.2023.10306450](https://doi.org/10.1109/icccnt56998.2023.10306450).
- [94] Y. Zhang, X. Zhang, and J. Du, "Non-invasive blood pressure measurement using a mobile phone camera," in *Proc. IEEE Conf. Comput. Intell. Bioinf. Comput. Biol. (CIBCB)*, vol. 10, pp. 1–6, Aug. 2022, doi: [10.1109/CIBCB55180.2022.9863015](https://doi.org/10.1109/CIBCB55180.2022.9863015).
- [95] S. Baek, J. Jang, S.-H. Cho, J. M. Choi, and S. Yoon, "Blood pressure prediction by a smartphone sensor using fully convolutional networks," in *Proc. 42nd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2020, pp. 188–191, doi: [10.1109/EMBC44109.2020.9175902](https://doi.org/10.1109/EMBC44109.2020.9175902).
- [96] M. Das, T. Choudhary, L. N. Sharma, and M. K. Bhuyan, "Noninvasive accelerometric approach for cuffless continuous blood pressure measurement," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–9, 2021, doi: [10.1109/TIM.2021.3122182](https://doi.org/10.1109/TIM.2021.3122182).
- [97] V. G. Ganti, A. M. Carek, B. N. Nevius, J. A. Heller, M. Etemadi, and O. T. Inan, "Wearable cuff-less blood pressure estimation at home via pulse transit time," *IEEE J. Biomed. Health Informat.*, vol. 25, no. 6, pp. 1926–1937, Jun. 2021, doi: [10.1109/JBHI.2020.3021532](https://doi.org/10.1109/JBHI.2020.3021532).
- [98] A. Zienkiewicz, E. Vihriälä, E. Seppälä, H. Ferdinando, and T. Mälylä, "Wearable sensor system on chest for continuous measurement of blood pressure and other vital signs," in *Proc. IEEE 16th Int. Symp. Med. Inf. Commun. Technol. (ISMICT)*, May 2022, pp. 1–6, doi: [10.1109/ISMICT56646.2022.9828307](https://doi.org/10.1109/ISMICT56646.2022.9828307).
- [99] J. Skoric et al., "Cuff-less estimation of blood pressure from vibrational cardiography using a convolutional neural network," in *Proc. Comput. Cardiology (CinC)*, vol. 498, Sep. 2022, pp. 1–4, doi: [10.22489/CinC.2022.110](https://doi.org/10.22489/CinC.2022.110).
- [100] Y. Zhang, X. Zhang, P. Cui, S. Li, and J. Tang, "Key feature selection and model analysis for blood pressure estimation from electrocardiogram, ballistocardiogram and photoplethysmogram," *IEEE Access*, vol. 9, pp. 54350–54359, 2021, doi: [10.1109/ACCESS.2021.3070636](https://doi.org/10.1109/ACCESS.2021.3070636).
- [101] S. Gjerde, T. Steffensen, H. Vestad, and M. Steinert, "Windows to the sole: Prototyping soft sensors for wearable ballistocardiography," in *Proc. IEEE-EMBS Int. Conf. Wearable Implant. Body Sensor Netw. (BSN)*, Sep. 2022, pp. 1–4, doi: [10.1109/BSN56160.2022.9928472](https://doi.org/10.1109/BSN56160.2022.9928472).
- [102] S.-H. Liu, B.-C. Lin, B.-H. Zhang, Y.-L. Hsu, and Y.-F. Huang, "The cuffless blood pressure measurement with ballistocardiogram and photoplethysmogram," in *Proc. IEEE Int. Conf. Consum. Electron.*, Jul. 2022, pp. 485–486, doi: [10.1109/ICCE-TAIWAN55306.2022.9869114](https://doi.org/10.1109/ICCE-TAIWAN55306.2022.9869114).
- [103] A. M. Carek, H. Jung, and O. T. Inan, "A reflective photoplethysmogram array and channel selection algorithm for weighing scale based blood pressure measurement," *IEEE Sensors J.*, vol. 20, no. 7, pp. 3849–3858, Apr. 2020, doi: [10.1109/JSEN.2019.2960063](https://doi.org/10.1109/JSEN.2019.2960063).
- [104] P. Fang et al., "Wrist pulse recording with a wearable piezoresistor-piezoelectret compound sensing system and its applications in health monitoring," *IEEE Sensors J.*, vol. 21, no. 18, pp. 20921–20930, Sep. 2021, doi: [10.1109/JSEN.2021.3094845](https://doi.org/10.1109/JSEN.2021.3094845).
- [105] A. Jinu and K. S. Biju, "Estimation of continuous blood pressure from pressure pulse wave morphology," in *Proc. Int. Conf. Control, Commun. Comput. (ICCC)*, May 2023, pp. 1–6, doi: [10.1109/ICCC57789.2023.10165042](https://doi.org/10.1109/ICCC57789.2023.10165042).
- [106] C.-Y. Guo, C.-C. Chang, K.-J. Wang, and T.-L. Hsieh, "Assessment of a calibration-free method of cuffless blood pressure measurement: A pilot study," *IEEE J. Transl. Eng. Health Med.*, vol. 11, pp. 318–329, 2023, doi: [10.1109/JTEHM.2022.3209754](https://doi.org/10.1109/JTEHM.2022.3209754).
- [107] Q. Zhang et al., "Highly sensitive resistance-type flexible pressure sensor for cuffless blood-pressure monitoring by using neural network techniques," *Compos. B, Eng.*, vol. 226, Dec. 2021, Art. no. 109365, doi: [10.1016/j.compositesb.2021.109365](https://doi.org/10.1016/j.compositesb.2021.109365).
- [108] F. Miao, Z.-D. Liu, J.-K. Liu, B. Wen, Q.-Y. He, and Y. Li, "Multi-sensor fusion approach for cuff-less blood pressure measurement," *IEEE J. Biomed. Health Informat.*, vol. 24, no. 1, pp. 79–91, Jan. 2020, doi: [10.1109/JBHI.2019.2901724](https://doi.org/10.1109/JBHI.2019.2901724).
- [109] C. Peng, M. Chen, H. K. Sim, Y. Zhu, and X. Jiang, "Noninvasive and nonocclusive blood pressure monitoring via a flexible piezo-composite ultrasonic sensor," *IEEE Sensors J.*, vol. 21, no. 3, pp. 2642–2650, Feb. 2021, doi: [10.1109/JSEN.2020.3021923](https://doi.org/10.1109/JSEN.2020.3021923).
- [110] C. Peng, M. Chen, H. K. Sim, Y. Zhu, and X. Jiang, "A flexible piezo-composite ultrasound blood pressure sensor with silver nanowire-based stretchable electrodes," in *Proc. IEEE 15th Int. Conf. Nano/Micro Engineered Mol. Syst. (NEMS)*, Sep. 2020, pp. 143–146, doi: [10.1109/NEMS50311.2020.9265560](https://doi.org/10.1109/NEMS50311.2020.9265560).
- [111] B. Jana, K. Oswal, S. Mitra, G. Saha, and S. Banerjee, "Windkessel model-based cuffless blood pressure estimation using continuous wave Doppler ultrasound system," *IEEE Sensors J.*, vol. 20, no. 17, pp. 9989–9999, Sep. 2020, doi: [10.1109/JSEN.2020.2990648](https://doi.org/10.1109/JSEN.2020.2990648).

- [112] I. Suriani et al., "Single-measurement local pulse wave velocity estimation in the carotid artery using pulsed wave Doppler ultrasound waveform indices," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–10, 2025, doi: [10.1109/TIM.2025.3571080](https://doi.org/10.1109/TIM.2025.3571080).
- [113] L. Xu et al., "Online continuous measurement of arterial pulse pressure and pressure waveform using ultrasound," *Measurement*, vol. 220, Oct. 2023, Art. no. 113379, doi: [10.1016/j.measurement.2023.113379](https://doi.org/10.1016/j.measurement.2023.113379).
- [114] D. Osman, M. Jankovic, K. Sel, R. I. Pettigrew, and R. Jafari, "Blood pressure estimation using a single channel bio-impedance ring sensor," in *Proc. 44th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2022, pp. 4286–4290, doi: [10.1109/EMBC48229.2022.9871653](https://doi.org/10.1109/EMBC48229.2022.9871653).
- [115] T.-W. Wang, W.-X. Chen, H.-W. Chu, and S.-F. Lin, "Single-channel bioimpedance measurement for wearable continuous blood pressure monitoring," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–9, 2021, doi: [10.1109/TIM.2020.3035578](https://doi.org/10.1109/TIM.2020.3035578).
- [116] B. Yang, L. Liu, W. Wu, M. Zhou, H. Liu, and X. Ning, "BrainZ-BP: A noninvasive cuff-less blood pressure estimation approach leveraging brain bio-impedance and electrocardiogram," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–13, 2023, doi: [10.1109/TIM.2023.3346527](https://doi.org/10.1109/TIM.2023.3346527).
- [117] M. Diehl, J. Zeilfelder, C. Zimmermann, and W. Stork, "Evaluation of a new method for continuous blood pressure measurement in the ear," in *Proc. IEEE Int. Symp. Medical Meas. Appl. (MeMeA)*, Jun. 2022, pp. 1–6, doi: [10.1109/MEMEA54994.2022.9856419](https://doi.org/10.1109/MEMEA54994.2022.9856419).
- [118] N. V. Kumar, S. Pant, S. Sridhar, V. Marulasiddappa, S. Srivatzen, and S. Asokan, "Fiber Bragg grating-based pulse monitoring device for real-time non-invasive blood pressure measurement—A feasibility study," *IEEE Sensors J.*, vol. 21, no. 7, pp. 9179–9185, Apr. 2021, doi: [10.1109/JSEN.2021.3055245](https://doi.org/10.1109/JSEN.2021.3055245).
- [119] Y. Koyama, H. Kamada, K. Watanabe, and M. Nishiyama, "Cuff-less blood pressure estimation using hetero-core optical fibers," *Electron. Commun. Jpn.*, vol. 107, no. 1, p. 1, Dec. 2023, doi: [10.1002/ecj.12438](https://doi.org/10.1002/ecj.12438).
- [120] H. Shi, J. Pan, Z. Zheng, B. Wang, C. Shen, and Y. Guo, "Radar-based blood pressure estimation using multiple features," in *Proc. IEEE MTT-S Int. Microwave Biomed. Conf.*, May 2022, pp. 183–185, doi: [10.1109/IMBIOC52515.2022.9790124](https://doi.org/10.1109/IMBIOC52515.2022.9790124).
- [121] H. Ochi, J. Liu, and S. Shimamoto, "Non-contact blood pressure estimation by microwave reflection employing machine learning," in *Proc. IEEE 19th Annu. Consum. Commun. Netw. Conf. (CCNC)*, Jan. 2022, pp. 889–892, doi: [10.1109/CCNC49033.2022.9700560](https://doi.org/10.1109/CCNC49033.2022.9700560).
- [122] Y. Ran, D. Zhang, J. Chen, Y. Hu, and Y. Chen, "Contactless blood pressure monitoring with mmWave radar," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2022, pp. 541–546, doi: [10.1109/GLOBECOM48099.2022.10001592](https://doi.org/10.1109/GLOBECOM48099.2022.10001592).
- [123] Z. Zheng, B. Wang, and Y. Guo, "Non-contact calibration-free blood pressure estimation method using dual radar," in *IEEE MTT-S Int. Microw. Symp. Dig.*, May 2022, pp. 186–188, doi: [10.1109/IMBIOC52515.2022.9790229](https://doi.org/10.1109/IMBIOC52515.2022.9790229).
- [124] H. Ochi, J. Liu, and S. Shimamoto, "Non-contact blood pressure prediction employing microwave reflection based on machine learning," in *Proc. IEEE 20th Consum. Commun. Netw. Conf. (CCNC)*, Jan. 2023, pp. 696–697, doi: [10.1109/CCNC51644.2023.10060068](https://doi.org/10.1109/CCNC51644.2023.10060068).
- [125] F. Geng et al., "Contactless and continuous blood pressure measurement according to caPTT obtained from millimeter wave radar," *Measurement*, vol. 218, Aug. 2023, Art. no. 113151, doi: [10.1016/j.measurement.2023.113151](https://doi.org/10.1016/j.measurement.2023.113151).
- [126] Y. Wang et al., "Radar-based contactless blood pressure estimation system using signal decomposition and deep neural network," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–12, 2025, doi: [10.1109/TIM.2025.3551902](https://doi.org/10.1109/TIM.2025.3551902).
- [127] S. Ishizaka, K. Yamamoto, and T. Ohtsuki, "Non-contact blood pressure measurement using Doppler radar based on waveform analysis by LSTM," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Jun. 2021, pp. 1–6, doi: [10.1109/ICC42927.2021.9500857](https://doi.org/10.1109/ICC42927.2021.9500857).
- [128] S. Ishizaka, K. Yamamoto, and T. Ohtsuki, "Non-contact blood pressure estimation method based on blood pressure category classification," in *Proc. 44th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2022, pp. 2676–2679, doi: [10.1109/EMBC48229.2022.9871918](https://doi.org/10.1109/EMBC48229.2022.9871918).
- [129] H. Ochi, S. Shimamoto, J. Liu, and Y. Yamaoka, "Non-contact blood pressure estimation with pulse wave employing microwave reflection," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, Jun. 2021, pp. 1–6, doi: [10.1109/ICCCWORKSHOPS50388.2021.9473654](https://doi.org/10.1109/ICCCWORKSHOPS50388.2021.9473654).
- [130] C. Wuerich, E.-M. Humm, C. Wiede, and G. Schiele, "A feature-based approach on contact-less blood pressure estimation from video data," in *Proc. 30th Eur. Signal Process. Conf. (EUSIPCO)*, Aug. 2022, pp. 1343–1347, doi: [10.23919/EUSIPCO55093.2022.9909563](https://doi.org/10.23919/EUSIPCO55093.2022.9909563).
- [131] B.-J. Wu, B.-F. Wu, and C.-P. Hsu, "Camera-based blood pressure estimation via Windkessel model and waveform features," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–13, 2023, doi: [10.1109/TIM.2022.3224534](https://doi.org/10.1109/TIM.2022.3224534).
- [132] B.-F. Wu, L.-W. Chiu, Y.-C. Wu, C.-C. Lai, and P.-H. Chu, "Contactless blood pressure measurement via remote photoplethysmography with synthetic data generation using generative adversarial network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 2129–2137, doi: [10.1109/CVPRW56347.2022.00231](https://doi.org/10.1109/CVPRW56347.2022.00231).
- [133] M. Li, X. Zhao, and J. Meng, "Non-contact monitoring of ambulatory blood pressure based on webcam," in *Proc. 34th Chin. Control Decis. Conf. (CCDC)*, Aug. 2022, pp. 1163–1167, doi: [10.1109/CCDC55256.2022.10033588](https://doi.org/10.1109/CCDC55256.2022.10033588).
- [134] F. Schruppf, P. Frenzel, C. Aust, G. Osterhoff, and M. Fuchs, "Assessment of deep learning based blood pressure prediction from PPG and rPPG signals," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 3815–3825, doi: [10.1109/CVPRW53098.2021.00423](https://doi.org/10.1109/CVPRW53098.2021.00423).
- [135] F. Shirbani et al., "Contactless video-based photoplethysmography technique comparison investigating pulse transit time estimation of arterial blood pressure," in *Proc. 43rd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Nov. 2021, pp. 5650–5653, doi: [10.1109/EMBC46164.2021.9629489](https://doi.org/10.1109/EMBC46164.2021.9629489).
- [136] C. Lumyong, N. Yodrabum, K. Winaikosol, and T. Titijaroonroj, "Skin video-based blood pressure approximation using CHROM with LSTM-NN," in *Proc. 15th Int. Conf. Knowl. Smart Technol. (KST)*, Feb. 2023, pp. 1–6, doi: [10.1109/KST57286.2023.10086816](https://doi.org/10.1109/KST57286.2023.10086816).
- [137] D. Qiao, A. H. Ayesha, F. Zulkernine, N. Jaffar, and R. Masroor, "ReViSe: Remote vital signs measurement using smartphone camera," *IEEE Access*, vol. 10, pp. 131656–131670, 2022, doi: [10.1109/ACCESS.2022.3229977](https://doi.org/10.1109/ACCESS.2022.3229977).
- [138] B.-F. Wu, B.-J. Wu, B.-R. Tsai, and C.-P. Hsu, "A facial-image-based blood pressure measurement system without calibration," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–13, 2022, doi: [10.1109/TIM.2022.3165827](https://doi.org/10.1109/TIM.2022.3165827).
- [139] R. H. Goudarz, S. S. Mousavi, and M. Charimi, "Introducing a new feature extraction method for non-contact blood pressure estimation through iPPG signals extracted using G-R method from video images of different facial regions," in *Proc. 5th Int. Conf. Pattern Recognit. Image Anal. (IPRIA)*, Apr. 2021, pp. 1–8, doi: [10.1109/IPRIA53572.2021.9483529](https://doi.org/10.1109/IPRIA53572.2021.9483529).
- [140] R. H. Goudarzi, S. S. Mousavi, and M. Charimi, "Using imaging photoplethysmography (iPPG) signal for blood pressure estimation," in *Proc. Int. Conf. Mach. Vis. Image Process. (MVIP)*, Feb. 2020, pp. 1–6, doi: [10.1109/MVIP49855.2020.9116902](https://doi.org/10.1109/MVIP49855.2020.9116902).
- [141] Y. Huang et al., "Camera-based blood pressure monitoring based on multisite and multiwavelength pulse transit time features," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–14, 2024, doi: [10.1109/TIM.2024.3457944](https://doi.org/10.1109/TIM.2024.3457944).
- [142] M. R. Shaikh, M. Rao, and G. Subramaniam, "A novel thermal imaging based transfer-learning model to estimate blood pressure," in *Proc. IEEE 20th Int. Symp. Biomed. Imag. (ISBI)*, Apr. 2023, pp. 1–5, doi: [10.1109/ISBI53787.2023.10230787](https://doi.org/10.1109/ISBI53787.2023.10230787).
- [143] Y. Iwashita, K. Nagumo, K. Oiwa, and A. Nozawa, "Evaluation of model performance for estimating resting blood pressure using independent components of facial thermal images," in *Proc. IEEE 19th Int. Symp. Intell. Syst. Informat. (SISY)*, Sep. 2021, pp. 135–140, doi: [10.1109/SISY52375.2021.9582441](https://doi.org/10.1109/SISY52375.2021.9582441).
- [144] X. Ding, W. Wang, Y. Chen, Y. Yang, Y. Zhao, and D. Kong, "Feasibility study of pulse width at half amplitude of camera PPG for contactless blood pressure estimation," in *Proc. 43rd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Nov. 2021, pp. 365–368, doi: [10.1109/EMBC46164.2021.9630964](https://doi.org/10.1109/EMBC46164.2021.9630964).
- [145] P. K. Parashiva, R. Damodaran, M. A. Kareem, and N. S. Narayan, "Device agnostic measurement of blood pressure from RPPG signals," in *Proc. 31st Eur. Signal Process. Conf. (EUSIPCO)*, Sep. 2023, pp. 1015–1019, doi: [10.23919/eusipco58844.2023.10289728](https://doi.org/10.23919/eusipco58844.2023.10289728).
- [146] C.-I. Kim, E.-B. Park, C.-H. Won, and J.-H. Lee, "Remote bio vision: Perfusion imaging based non-contact autonomic biosignal measurement algorithm," *IEEE Sensors J.*, vol. 23, no. 14, pp. 16324–16331, Jul. 2023, doi: [10.1109/JSEN.2023.3279328](https://doi.org/10.1109/JSEN.2023.3279328).
- [147] S. Subramani, S. Sukumar, Z. Ahmed, V. Venkat, V. Sriram, and B. Geethanjali, "Analysing the performance of blood pressure parameters using EMF method," in *Proc. 4th Int. Conf. Trends Electron. Informat. (ICOEI)*(4), Jun. 2020, pp. 832–836, doi: [10.1109/ICOEI48184.2020.9142931](https://doi.org/10.1109/ICOEI48184.2020.9142931).

- [148] Y. Oyamada et al., "Laser-based noncontact blood pressure estimation using human body displacement waveforms," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Jun. 2022, pp. 1020–1022, doi: [10.1109/IMS37962.2022.9865553](https://doi.org/10.1109/IMS37962.2022.9865553).
- [149] T. Vajanarat and A. Lek-uthai, "A comparison of cuff-less blood pressure estimation between pulse arrival time and pulse transit time using photoplethysmography," in *Proc. 17th Int. Conf. Electr. Eng./Electron., Comput., Telecommun. Inf. Technol. (ECTI-CON)*, Jun. 2020, pp. 13–16, doi: [10.1109/ECTI-CON49241.2020.9158100](https://doi.org/10.1109/ECTI-CON49241.2020.9158100).
- [150] A. Ghosh, T. Chatterjee, and S. Sarkar, "Introduction of boosting algorithms in continuous non-invasive cuff-less blood pressure estimation using pulse arrival time," in *Proc. 43rd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Nov. 2021, pp. 5429–5432, doi: [10.1109/EMBC46164.2021.9630848](https://doi.org/10.1109/EMBC46164.2021.9630848).
- [151] S. Kiliçkaya, A. Güner, and B. Dal, "Comparison of different machine learning techniques for the cuffless estimation of blood pressure using PPG signals," in *Proc. Int. Congr. Hum.-Comput. Interact., Optim. Robotic Appl. (HORA)*, Jun. 2020, pp. 1–6, doi: [10.1109/HORA49412.2020.9152602](https://doi.org/10.1109/HORA49412.2020.9152602).
- [152] W. Shi et al., "Non-invasive blood pressure monitoring using a single-channel PPG sensor with adaptive Kalman algorithm and 4-LED arrayed structure," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–14, 2024, doi: [10.1109/TIM.2024.3480192](https://doi.org/10.1109/TIM.2024.3480192).
- [153] M. Caillat et al., "Accuracy of blood pressure measurement across BMI categories using the OptiBP™ mobile application," *Blood Pressure*, vol. 31, no. 1, pp. 288–296, Dec. 2022, doi: [10.1080/08037051.2022.2132214](https://doi.org/10.1080/08037051.2022.2132214).
- [154] M. M. R. K. Mamun and A. T. Alouani, "Cuffless blood pressure measurement using linear and nonlinear optimized feature selection," *Diagnostics*, vol. 12, no. 2, p. 408, Feb. 2022, doi: [10.3390/diagnostics12020408](https://doi.org/10.3390/diagnostics12020408).
- [155] U. C. Akuthota, M. M. Hassan, and L. Bhargava, "Blood pressure prediction based on single photoplethysmography," in *Proc. IEEE Int. Symp. Smart Electron. Syst. (iSES)*, Dec. 2022, pp. 337–342, doi: [10.1109/ISES54909.2022.00075](https://doi.org/10.1109/ISES54909.2022.00075).
- [156] R. Moreno et al., "IoT medical device architecture to estimate non-invasive arterial blood pressure," in *Proc. Symp. Internet Things (SIoT)*, Oct. 2022, pp. 1–4, doi: [10.1109/SIoT56383.2022.10069878](https://doi.org/10.1109/SIoT56383.2022.10069878).
- [157] W. Wang, L. Zhu, F. Marefat, P. Mohseni, K. Kilgore, and L. Najafizadeh, "Photoplethysmography-based blood pressure estimation using deep learning," in *Proc. 54th Asilomar Conf. Signals, Syst., Comput.*, Nov. 2020, pp. 945–949, doi: [10.1109/IEEECONF51394.2020.9443447](https://doi.org/10.1109/IEEECONF51394.2020.9443447).
- [158] E. A. Aldein, M. Abdlerahem, U. S. Mohamed, and M. Atef, "An ECG-based blood pressure estimation using U-Net auto-encoder and random forest regressor," in *Proc. Int. Conf. Microelectron. (ICM)*, Dec. 2023, pp. 107–112, doi: [10.1109/icm60448.2023.10378899](https://doi.org/10.1109/icm60448.2023.10378899).
- [159] C. El Hajj and P. A. Kyriacou, "Cuffless and continuous blood pressure estimation from PPG signals using recurrent neural networks," in *Proc. 42nd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2020, pp. 4269–4272, doi: [10.1109/EMBC44109.2020.9175699](https://doi.org/10.1109/EMBC44109.2020.9175699).
- [160] C. Ma et al., "KD-informer: A cuff-less continuous blood pressure waveform estimation approach based on single photoplethysmography," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 5, pp. 2219–2230, May 2023, doi: [10.1109/JBHI.2022.3181328](https://doi.org/10.1109/JBHI.2022.3181328).
- [161] G. S. Stergiou et al., "A universal standard for the validation of blood pressure measuring devices: Association for the advancement of medical instrumentation/European society of hypertension/International organization for standardization (AAMI/ESH/ISO) collaboration statement," *Hypertension*, vol. 71, no. 3, pp. 368–374, Mar. 2018, doi: [10.1161/hypertensionaha.117.10237](https://doi.org/10.1161/hypertensionaha.117.10237).
- [162] Y. Chu et al., "Non-invasive arterial blood pressure measurement and SpO₂ estimation using PPG signal: A deep learning framework," *BMC Med. Informat. Decis. Making*, vol. 23, no. 1, pp. 1–16, Jul. 2023, doi: [10.1186/s12911-023-02215-2](https://doi.org/10.1186/s12911-023-02215-2).
- [163] C.-H. Tseng, T.-J. Tseng, and C.-Z. Wu, "Cuffless blood pressure measurement using a microwave near-field self-injection-locked wrist pulse sensor," *IEEE Trans. Microw. Theory Techn.*, vol. 68, no. 11, pp. 4865–4874, Nov. 2020, doi: [10.1109/TMTT.2020.3011446](https://doi.org/10.1109/TMTT.2020.3011446).
- [164] M. P. D. Pont and J. L. B. Marques, "Reflective photoplethysmography acquisition platform with monitoring modules and noninvasive blood pressure calculation," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 8, pp. 5649–5657, Aug. 2020, doi: [10.1109/TIM.2019.2963508](https://doi.org/10.1109/TIM.2019.2963508).
- [165] R. Maharajan, "Cuffless BP measurement using single site photoplethysmography," in *Proc. Int. Conf. Bio Signals, Images, Instrum. (ICBSII)*, Mar. 2023, pp. 1–5, doi: [10.1109/icbsii58188.2023.10181082](https://doi.org/10.1109/icbsii58188.2023.10181082).
- [166] H. Samimi and H. R. Dajani, "Cuffless blood pressure estimation using cardiovascular dynamics," in *Proc. Int. Conf. Electr., Comput. Energy Technol. (ICECET)*, Jul. 2022, pp. 1–8, doi: [10.1109/ICECET55527.2022.9872800](https://doi.org/10.1109/ICECET55527.2022.9872800).
- [167] T. Panula, J.-P. Sirkkiä, D. Wong, and M. Kaisti, "Advances in non-invasive blood pressure measurement techniques," *IEEE Rev. Biomed. Eng.*, vol. 16, pp. 424–438, 2023, doi: [10.1109/RBME.2022.3141877](https://doi.org/10.1109/RBME.2022.3141877).
- [168] R. Jegan, K. M. Sneha, and W. S. Nimi, "Review of non-invasive blood pressure estimation via modern approaches," in *Proc. 3rd Int. Conf. Smart Electron. Commun. (ICOSEC)*, Oct. 2022, pp. 874–878, doi: [10.1109/ICOSEC54921.2022.9952014](https://doi.org/10.1109/ICOSEC54921.2022.9952014).
- [169] S. Manga et al., "Estimation of physiologic pressures: Invasive and non-invasive techniques, AI models, and future perspectives," *Sensors*, vol. 23, no. 12, p. 5744, Jun. 2023, doi: [10.3390/s23125744](https://doi.org/10.3390/s23125744).
- [170] Non-Invasive Sphygmomanometers—Part 2: Clinical Investigation of Intermittent Automated Measurement Type, ISO Standard 81060-2:2018, 2018. [Online]. Available: <https://www.iso.org/standard/73339.html>
- [171] Non-Invasive Sphygmomanometers—Part 3: Clinical Investigation of Continuous Automated Measurement Type, ISO Standard 81060-3:2022, 2022. [Online]. Available: <https://www.iso.org/standard/71161.html>
- [172] Wearable Cuffless Blood Pressure Measuring Devices, IEEE Standard 1708-2014, 2014. [Online]. Available: <https://standards.ieee.org/ieee/1708/5927/>
- [173] Amendment 1: Wearable Cuffless Blood Pressure Measuring Devices, IEEE Standard 1708a-2019, 2019. [Online]. Available: <https://standards.ieee.org/ieee/1708a/7336/>
- [174] Medical Devices—Application of Risk Management To Medical Devices, ISO Standard 14971:2019, 2019, p. 2019. [Online]. Available: <https://www.iso.org/standard/72704.html>
- [175] Information Security, Cybersecurity and Privacy Protection—Information Security Management Systems—Requirements, ISO/IEC Standard 27001:2022, 2022. [Online]. Available: <https://www.iso.org/standard/27001>
- [176] Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the Protection of Natural Persons With Regard to the Processing of Personal Data and on the Free Movement of Such Data, and Repealing Directive 95/46/EC (General Data Protection Regulation), Off. J. Eur. Union, Brussels, Belgium, May 2016, vol. L119, pp. 1–88.
- [177] Clinical Investigation of Medical Devices for Human Subjects—Good Clinical Practice, ISO Standard 14155:2020, 2020. [Online]. Available: <https://www.iso.org/standard/71690.html>